

GROUP 5 - BACHELOR PROJECT FINAL REPORT

Numerical Modelling Tool for Load Sharing Behaviour of Epicyclic Gears

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Date: May 2026
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1. Introduction

Background and Problem

Planetary gear systems, also known as epicyclic gear systems, are one of the most widely used power transmission systems in mechanical engineering. They are used in many real-world applications, including wind turbine gearboxes, helicopter main rotor drives, and turboprop engines. As highlighted by Gu and Velez [4], planetary gears offer coaxial shafts, high power density and high mechanical efficiency, making them particularly suitable for compact and lightweight transmission designs.

One of the main reasons for their high power density is the load-sharing capability between multiple planet gears. In a conventional two-gear system, the entire input torque is transmitted through a single meshing contact. On the other hand, in a planetary gear system, the input torque is divided among multiple parallel paths. For an ideal system with N planet gears, each planet gear ideally carries exactly $1/N$ of the total input torque.

In practice, this perfect distribution is rarely achieved. Real systems contain manufacturing errors, misalignments, and stiffness variations which cause each planet to end up carrying a different portion of the total load. This load-sharing imbalance may eventually damage the system by increasing local stresses and accelerating wear, causing the gearbox behaviour to gradually deviate from the ideal load-sharing condition. The purpose of the numerical tool developed in this project is therefore to study how various parameters and effects influence load-sharing behaviour in epicyclic gear systems while reducing the necessity for manual or analytical post-processing.

Project Objectives

The objective of this Final Project, as defined in the original project brief, was to develop an analytical tool capable of estimating unequal load sharing between planet gears. The model was intended to consider geometric defects, mechanical stiffness, and sun gear orbital motion in order to predict load distribution within epicyclic gear systems.

As the project developed, the tool evolved from a simplified load-sharing calculation into a numerical simulation model of coupled force redistribution between the planet gears. In the final version, the model includes planet-by-planet force prediction, floating sun displacement and tilt, phase-dependent behaviour, and optional sensitivity and Monte Carlo analyses. The overall objective was therefore expanded from calculating isolated

scalar values toward studying the global behaviour of planetary gear systems under non-ideal operating conditions. At the same time, the number of user-defined inputs was intentionally kept as low as possible in order to balance physical fidelity, usability, and computational efficiency.

Development and Progression

The initial version of the numerical modelling tool was based primarily on the Epicyclic Load Sharing Map (ELSM) concept developed by A. Singh [2] and later refined using the Translational Analogy (TA) framework. In this early formulation, the planetary gear set was represented using simplified lumped-parameter springs and rigid bodies. Mesh stiffness was treated as linear and constant, manufacturing errors were introduced as fixed or random geometric offsets, and the sun gear displacement was solved using the resultant forces generated by the surrounding planet branches.

Although this approach reproduced accurate results compared to literature cases, it had important limitations. The planet contacts were not fully coupled, stiffness modelling remained highly simplified, and the orbital motion of the sun gear was restricted to planar translation only. In addition, the original interface functioned mainly as a command-line calculation tool and provided only limited output quantities.

As development progressed, the project moved away from reproducing isolated benchmark calculations and toward building a more flexible and physically consistent equilibrium model. The final solver architecture treats the planet branches as a connected system coupled through the generalized displacement and tilt coordinates of the floating sun gear. This allows local stiffness variations or geometric errors at one planet to influence the equilibrium position of the sun and redistribute the contact forces across the remaining planets. These changes also expanded the functionality of the numerical tool itself – the final version includes phase-dependent effects, thermal mismatch modelling, plot visualization, and a graphical user interface designed to simplify parameter studies and result interpretation across different operating conditions and system configurations.

2. Geometry

System and Gear Configurations

The numerical tool was developed to account for the main geometrical parameters influencing load-sharing behaviour in planetary gear systems. These parameters are directly linked to the reduced-order equations used in the final model and therefore define the geometrical quantities that can be assigned by the user through the graphical interface. The following sections introduce these parameters and explain their physical meaning within the context of real epicyclic gear systems before their mathematical formulation is developed later in the report.

A planetary gear system consists of four main components, illustrated in the Figure 2.1 below: the sun gear, located at the centre of the mechanism and typically connected to the input shaft; the planet gears, which rotate around the sun gear; the ring gear, an outer annular gear with internal teeth sharing the same central axis as the sun gear; and the planet carrier, which supports and positions the planet gears within the system.

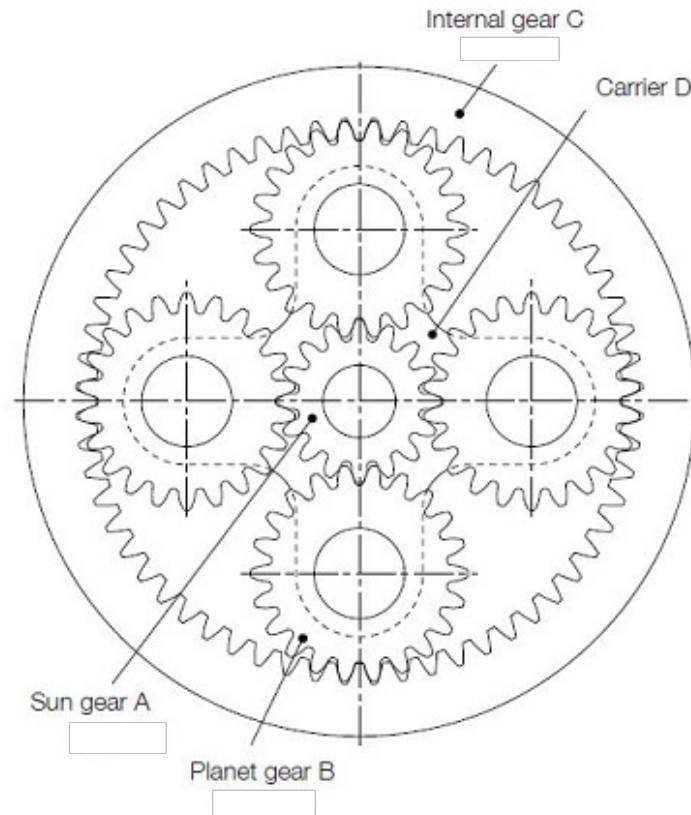


Figure 2.1: Primary components of a planetary gear system

System Parameters

The main parameters defining the configuration of the planetary gear system are the number of planets N , their angular positions ψ_i , and the tooth numbers of the sun, planet, and ring gears (referenced as z_1 , z_2 , and z_3 respectively). These parameters determine the geometrical arrangement of the system and directly influence the load-sharing behaviour predicted by the model. For an ideal system containing N identical planet gears, the input load is equally distributed between all planet branches. The ideal force carried by each planet is therefore:

$$F_{\text{ideal}} = \frac{W_{\text{nominal}}}{N} = \frac{T_{\text{input}}}{N \cdot R_{\text{sun}}}$$

where T_{input} is the applied input torque and R_{sun} is the pitch radius of the sun gear. In practice, geometric and stiffness variations cause the load carried by each planet to deviate from this ideal condition.

The number of planets influences both the nominal load division and the overall equilibrium of the floating sun gear. Increasing the number of planets introduces additional contact branches, each contributing to the redistribution of forces and geometric compatibility within the system. By default, the planets are assumed to be equally spaced around the sun gear according to the standard symmetric configuration:

$$\psi_i = (i - 1) \frac{2\pi}{N} \quad \text{for } i = 1, \dots, N$$

Non-uniform custom angular spacing can also be defined to represent asymmetric layouts or assembly deviations, which may alter the equilibrium position and orbital motion of the floating sun gear. The tooth numbers of the sun, planet, and ring gears are also included because they influence several geometrical quantities used in the model, particularly the mesh stiffness calculations introduced later in the report. For simplified theoretical studies, the model can alternatively use predefined reference gear geometries representing a standard baseline configuration.

Gear Macro-Geometry

The model also includes several macro-geometrical parameters which define the gear meshes and contact conditions within the planetary system. These include the module (m_0), helix angle (β), pressure angle (α_n), working centre distance (a_w), and face width (b). These parameters are retained because they directly influence two important parts of the numerical model: the line-of-action (LOA) direction used for geometric error projection and the mesh stiffness formulation used in the equilibrium solver.

If no custom operating pressure angle is specified, the operating pressure angle along the contact line is derived from the normal pressure angle and helix angle using:

$$\tan(\alpha_w) = \frac{\tan(\alpha_n)}{\cos(\beta)}$$

The module, face width, pressure angle, and gear tooth geometry also influence the transverse contact ratio and, consequently, the estimated mesh stiffness bounds used to construct the baseline mean mesh stiffness of the system. These stiffness formulations are developed in later sections of the report.

Real Errors

Real planetary gear systems are affected by many sources of geometric and mechanical deviation. These include manufacturing tolerances, assembly inaccuracies, support misalignment, thermal expansion, eccentricity, and elastic deformation under load. Such imperfections disturb the ideal symmetry of the gearbox and modify how the transmitted load is distributed between the planet gears. In any reduced-order model, not all physical defects can be represented individually – the final numerical tool therefore retains only the error categories that can be meaningfully incorporated into the compatibility and equilibrium framework used by the solver.

These errors are particularly important because even relatively small geometric deviations can significantly disturb the function and longevity of a system. When one planet gear carries a disproportionately large share of the transmitted load, the resulting local stress concentration may accelerate wear, increase fatigue damage, and reduce the operational lifespan of the gearbox. The purpose of the numerical tool is therefore not only to calculate unequal load distribution itself, but also to provide a simplified engineering representation of the physical mechanisms that may lead to long-term mechanical degradation in real planetary gear systems.

Common Error Categories

The main error categories considered in the final model are:

- pin-position errors, including radial and tangential offsets;
- planet runout and assembly eccentricity;
- tooth profile errors;
- rigid assembly offsets, such as sun or carrier displacement;
- translational x, y offsets;

- thermal expansion mismatch;
- phase-dependent periodic eccentricity effects.

These error sources were selected because they directly influence either the compatibility conditions between planet branches, the equilibrium position of the floating sun gear, or the resulting load-sharing distribution. Other physical effects that are not directly associated to having an effect on these factors – such as surface roughness, lubricant viscosity, or the possibility of system contamination by dirt or metal debris – were not included in the final model nor the reduced-order equilibrium formulation.

Line-of-Action Projection

Many real geometric deviations in planetary gear systems are naturally two-dimensional. For example, a pin-position error or assembly offset may contain both x - and y -components. However, the reduced-order solver does not use the full displacement vector directly. Instead, each geometric deviation is projected onto the line of action (LOA), which is the direction along which the contact force is transmitted between meshing gear teeth.

This projection is important because only the component of a displacement acting along the line of action directly affects local mesh compatibility. A large displacement occurring mostly perpendicular to the contact direction may have little influence on load sharing, while a smaller displacement aligned with the line of action may strongly affect planet engagement and force redistribution. The purpose of the LOA formulation is therefore to convert real geometric deviations into a scalar compatibility quantity that can be used within the equilibrium solver.

In the model, the line-of-action direction for planet i is defined from the planet angular position ψ_i and the operating pressure angle α_w :

$$\theta_{\text{LOA},i} = \psi_i - \alpha_w + \frac{\pi}{2}$$

The corresponding unit direction vector is then written as:

$$\mathbf{n}_{\text{LOA},i} = \begin{bmatrix} \cos(\theta_{\text{LOA},i}) \\ \sin(\theta_{\text{LOA},i}) \end{bmatrix}$$

where $\mathbf{n}_{\text{LOA},i}$ is used to project the relevant displacement of planet i onto its local contact direction. This projection step allows the compatibility conditions of all planet branches to be represented using a single scalar equivalent error value for each planet, e_i . A schematic representation of the line of action is shown in Figure 2.2.

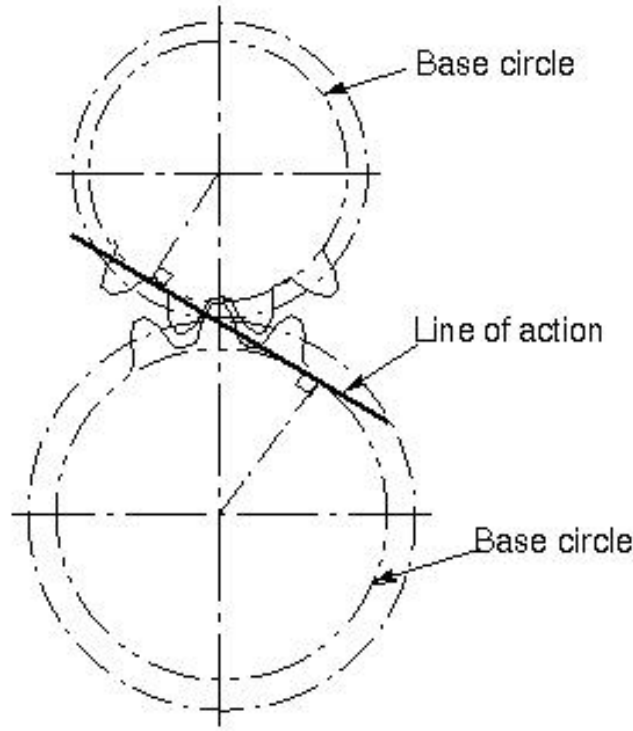


Figure 2.2: Line-of-action schematic for reference

Equivalent Errors

The equivalent error represents the effective contact mismatch produced by the real geometric deviations acting on a given planet branch. In the reduced-order model, it is defined as the projection of the local displacement vector onto the line-of-action direction:

$$e_i = -\mathbf{u}_i \cdot \mathbf{n}_{\text{LOA},i}$$

where $\mathbf{u}_i$ represents the local two-dimensional geometric deviation associated with planet i , and $\mathbf{n}_{\text{LOA},i}$ is the corresponding line-of-action unit vector. Physically, the equivalent error describes whether a planet is effectively advanced or delayed in contact relative to the surrounding planet branches. A positive or negative equivalent error therefore changes the local compatibility condition and influences how the transmitted load is redistributed within the system.

The solver does not resolve detailed tooth geometry or local contact deformation directly. Instead, all relevant geometric deviations must first be converted into this common scalar compatibility quantity before the equilibrium problem can be solved. The equivalent-error representation therefore provides the mathematical link between real physical imperfections and the compatibility-gap formulation used by the numerical model.

Gap Contact Model

Once the equivalent errors e_i have been obtained for each planet branch, the solver must determine which planets are in active contact and which remain separated by a clearance gap. Planets with smaller compatibility gaps engage contact earlier and therefore begin carrying load sooner, while planets with larger gaps engage later and contribute less to the total transmitted force. Unequal engagement between planet branches therefore leads directly to load-sharing imbalance.

To represent this behaviour, the equivalent errors are converted into compatibility gaps δ_i , defined relative to the most protruding planet branch:

$$\delta_i = e_{\max} - e_i \quad \text{where} \quad e_{\max} = \max_j(e_j)$$

This formulation guarantees that $\delta_i \geq 0$ for all planets. The planet with $\delta_i = 0$ becomes the first branch to engage contact, while the remaining planets must elastically close their compatibility gaps before carrying load. Larger values of δ_i therefore correspond to delayed engagement and reduced participation in load sharing.

The compatibility-gap formulation forms the basis of the contact-resolution procedure used by the numerical solver. Through this procedure, the solver determines the floating sun gear displacement and tilt required to satisfy global equilibrium while simultaneously resolving the compression-only contact forces acting at each planet branch. As δ_i increases, planet i is progressively unloaded relative to its ideal share.

3. Error Framework

Static Errors

After defining the equivalent line-of-action error framework, the final model constructs the static error of each planet as the sum of several manufacturing and assembly contributions. These errors are called static because they remain fixed throughout a phase sweep and therefore represent the baseline geometric state of the gearbox before any phase-dependent effects are introduced.

For each planet i , the static equivalent error is assembled as:

$$e_{\text{static},i} = \sum_k e_{k,i}$$

In the present implementation, the principal contributions are pin-position, runout, rigid-offset and profile errors. Each term represents a different way in which the real gearbox can lose perfect geometric symmetry. All static error sources are ultimately expressed as equivalent line-of-action compatibility errors so that their influence on planet engagement and load sharing can be treated within a common equilibrium framework.

Pin-Position Errors

Pin-position error represents the deviation of a planet pin from its ideal location. In the final model, this error is divided into two local components, a radial component Δr_{pin} and a tangential component Δt_{pin} . These two components are first expressed as a two-dimensional displacement vector using the local radial and tangential directions of the planet. This gives:

$$\mathbf{u}_{\text{pin},i} = \Delta r_{\text{pin},i} \mathbf{e}_{R,i} + \Delta t_{\text{pin},i} \mathbf{e}_{T,i}$$

Thus:

$$e_{\text{pin},i} = -\mathbf{u}_{\text{pin},i} \cdot \mathbf{n}_{\text{LOA},i}$$

The resulting equivalent error depends on both the magnitude and orientation of the pin-position deviation relative to the local contact direction.

Planet Runout

Planet runout represents an offset between the planet gear's geometric centre and its axis of rotation. In the static error model, runout is treated as a fixed local eccentricity

contribution for each planet. It is represented by an amplitude and an angular direction, giving a two-dimensional displacement vector:

$$\mathbf{u}_{\text{runout},i} = \begin{bmatrix} A_{\text{runout},i} \cos(\varphi_i) \\ A_{\text{runout},i} \sin(\varphi_i) \end{bmatrix}$$

$$e_{\text{runout},i} = -\mathbf{u}_{\text{runout},i} \cdot \mathbf{n}_{\text{LOA},i}$$

This static runout term should be distinguished from the optional periodic eccentricity term discussed later. Static runout belongs to the fixed manufacturing error realization, while periodic eccentricity is a rotating contribution added during the phase sweep.

Rigid Assembly Offsets

Rigid assembly offsets represent global misalignments between system reference positions, such as the sun centre, carrier centre, common assembly displacement or additional relative offsets. These offsets do not represent tooth profile defects directly; instead, they shift the relative positions of the components that define the planet branch compatibility.

$$e_{\text{rigid},i} = -\mathbf{u}_{\text{rigid},i} \cdot \mathbf{n}_{\text{LOA},i}$$

In the model, the rigid offset contribution is formed from the relevant system-level offsets and projected such that $\mathbf{u}_{\text{rigid},i}$ represents the combined rigid displacement affecting planet i . This term allows the model to include global assembly effects without requiring a full geometric reconstruction of the gearbox.

Profile & Common Profile Errors

Profile error represents a local deviation in the tooth profile that changes the effective contact condition of a planet branch. Because the model does not resolve tooth-level contact geometry, profile error is treated directly as an equivalent scalar contribution.

The model distinguishes between individual profile error, which may differ from planet to planet and common profile error, which is applied in the same way across planets. The corresponding contribution is written as:

$$e_{\text{profile},i} = e_{\text{individual profile},i} + e_{\text{common profile}}$$

Static errors influence the load-sharing behaviour of the gearbox through their effect on contact compatibility between planet branches. Differences in the equivalent error distribution modify the compatibility gaps of individual planets, alter the equilibrium position of the floating sun gear, and ultimately change the force carried by each planet

branch. Although the model does not attempt to reproduce every microscopic imperfection, the static error framework captures the dominant manufacturing and assembly deviations responsible for unequal load sharing.

Tolerance Bound Models

The numerical tool includes three tolerance presets representing different manufacturing-quality levels. These presets provide representative micron-level error magnitudes, or bounds, that can be used to investigate how variations in manufacturing and assembly accuracy influence load-sharing behaviour. The three presets are defined as:

- **Fine:** precision-manufactured or more accurate case with lower initial mismatches;
- **Medium:** standard engineering case used as the default baseline;
- **Coarse:** less precise or intentionally degraded case with larger assumed geometric deviations.

The tables below (Tables 3.1, 3.2, 3.3) specify the exact values that were chosen for each error category of the three bounds.

Table 3.1: Fine tolerance preset used for static error generation.

Error category	Tolerance value [μm]
Planet pin radial error	8
Planet pin tangential error	8
Planet runout	5
Tooth profile error	4
Common profile error	2
Common XY assembly offset	2

Table 3.2: Medium tolerance preset used for static error generation.

Error category	Tolerance value [μm]
Planet pin radial error	15
Planet pin tangential error	15
Planet runout	10
Tooth profile error	8
Common profile error	3
Common XY assembly offset	3

Table 3.3: Coarse tolerance preset used for static error generation.

Error category	Tolerance value [μm]
Planet pin radial error	25
Planet pin tangential error	25
Planet runout	18
Tooth profile error	12
Common profile error	5
Common XY assembly offset	5

Limitations of the Preset Approach

These presets do not represent formal manufacturing standards, however. ISO 1328 and AGMA tolerance limits were not used directly because their allowable deviations depend on gear size, geometry, and quality grade, whereas these quantities may vary between simulations and are not explicitly defined as user inputs. This is why representative micron-level deviations were therefore adopted instead. The selected values lie within the single-digit to low-tens-of-microns range, providing realistic error magnitudes for planetary load-sharing, sensitivity, and Monte Carlo studies without requiring a formal tolerance classification.

Stochastic Error Modelling

Manufacturing and assembly errors are inherently variable and therefore cannot be represented adequately by a single deterministic error realization. Even gearboxes produced from the same nominal design will contain different combinations of geometric deviations, leading to different load-sharing responses. For this reason, the numerical tool includes a stochastic error framework that generates multiple error realizations within the selected tolerance limits.

Random error generation is applied to the static manufacturing and assembly error components, including pin-position errors, runout, profile errors, and common assembly offsets. The selected tolerance values are first converted from microns to metres and then used to generate the corresponding equivalent line-of-action errors. The generated values may be sampled from either a normal distribution or a uniform distribution. These two options represent different assumptions about how manufacturing errors are distributed: a normal distribution assumes that smaller deviations are more likely than larger ones, whereas a uniform distribution assumes that all values within the tolerance interval are equally likely.

The tolerance value can be interpreted in two ways. If `tolMeaning = sigma`, the selected tolerance is treated directly as the standard deviation of the sampled error. If `tolMeaning = pm`, the tolerance is interpreted as a plus/minus manufacturing bound:

$$-T \leq e \leq T$$

For normally distributed errors in the plus/minus case, the model converts this bound into an equivalent standard deviation using:

$$\sigma = \frac{T}{\text{sigmaRule}}$$

In the default configuration, `sigmaRule = 3`, meaning that the tolerance interval is treated approximately as a three-sigma range. If truncation is enabled, generated values outside the specified tolerance bounds are rejected and resampled before being used in the analysis. For each realization, the complete equivalent-error vector is assembled and the planetary equilibrium problem is solved. Repeating this procedure over multiple realizations allows the influence of manufacturing uncertainty on load-sharing behaviour to be assessed. Larger tolerance values generally produce larger compatibility errors, greater force redistribution, and increased deviation from the ideal load-sharing condition. The resulting distributions of quantities such as the maximum load-sharing index, planet forces, and floating-sun displacement therefore provide an indication of the sensitivity and robustness of the system to manufacturing variation.

Thermal Geometry Mismatch

An additional thermal-mismatch model was included to investigate how non-uniform temperature distributions may influence load sharing. The key modelling assumption is that a perfectly uniform temperature increase should not, by itself, create significant load imbalance. If all components expand uniformly, most of that expansion is common-mode and does not significantly change which planet is more advanced or delayed relative to the others. For this reason, the thermal mismatch model separates two physical mechanisms. The first represents thermal expansion of the carrier and pin-circle geometry, which alters the planet positions. The second represents thermal growth of the planet gears themselves, which changes the effective contact geometry. Both effects are converted into equivalent line-of-action errors and combined within the compatibility framework.

To generate differential thermal expansion between planets, the model focuses on temperature variations around the gearbox rather than on the mean temperature rise alone. The underlying idea is that some regions of a gearbox may operate at slightly different temperatures due to local loading, frictional heating, lubrication conditions,

or cooling effects. These temperature differences produce unequal thermal expansion between planet branches, which can modify the compatibility conditions responsible for load sharing. To represent this behaviour, a circumferential temperature variation is imposed around the planetary system:

$$\Delta T_i = \Delta T_{\text{mean}} + \Delta T_{\text{grad}} \cos(\psi_i - \psi_{\text{hot}})$$

where ΔT_{mean} represents the average temperature rise of the gearbox, ΔT_{grad} defines the amplitude of the thermal gradient, and ψ_{hot} specifies the angular location of the hottest region. If the automatic thermal gradient option is used, the gradient is calculated as a fraction of the mean temperature rise such that:

$$\Delta T_{\text{grad}} = r_{\text{grad}} |\Delta T_{\text{mean}}|$$

where r_{grad} is the thermal gradient ratio, defaulted in the code as a value of 0.10. This formulation creates a smooth temperature field in which different planets experience slightly different local temperatures. A cosine distribution was selected because it provides a smooth circumferential temperature field while requiring only three parameters: the mean temperature rise, gradient magnitude, and hotspot location. This allows thermal asymmetry to be represented without introducing a detailed thermal model.

The resulting temperature differences generate two reduced-order thermal geometry effects. The first represents thermal expansion of the carrier and planet-pin locations:

$$\mathbf{u}_{\text{pin,th},i} = \alpha a_w \Delta T_i \mathbf{e}_{R,i}$$

with $\mathbf{e}_{R,i}$ being the radial unit vector pointing from the centre of the system toward planet i , and α being the linear thermal expansion coefficient. In the final model, the default value corresponds to a generic ferrous alloy / steel approximation: $\alpha = 11.5 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$. The resultant thermal radial displacement vector of the planet pin location is subsequently projected onto the line of action to obtain an equivalent compatibility error.

The second equation represents the effective thermal growth of the planet gear itself:

$$e_{\text{planet,th},i} = \alpha r_{\text{planet}} \Delta T_i \sin(\alpha_p)$$

where r_{planet} is the planet pitch radius, and the pressure angle used in the line-of-action model is denoted by the symbol α_p . The two contributions are then combined into a single thermal equivalent error:

$$e_{\text{th},i} = w_{\text{pin}} e_{\text{pin,th},i} + w_{\text{planet}} e_{\text{planet,th},i}$$

The weighting factors of this equation ($w_{\text{pin}}, w_{\text{planet}}$) were selected empirically to balance the relative influence of carrier expansion and planet thermal growth within the reduced-order framework (set at values of 1.0 and 0.7, respectively). Because the objective is to represent relative thermal mismatch rather than uniform expansion, the mean thermal contribution is removed:

$$e_{\text{th},i} \leftarrow e_{\text{th},i} - \overline{e_{\text{th}}}$$

This prevents a uniform temperature rise from artificially generating load redistribution or altering relative load sharing, as the model is intended to capture differential thermal mismatch rather than common-mode thermal growth. The resulting thermal mismatch error is then added to the equivalent-error framework and treated in the same manner as the other geometric error sources. Thermal effects are therefore represented using a reduced-order engineering approximation suitable for comparative studies. The model captures the influence of planet-to-planet thermal mismatch on load sharing, but it should not be interpreted as a full thermo-elastic contact or finite-element thermal analysis.

4. Stiffness Framework

Role of Gear Compliance

Stiffness is a central factor in load sharing because geometric errors alone do not directly determine how forces are distributed within a mechanical system. Instead, these errors create compatibility mismatches between the components, while the stiffness of each branch determines how strongly these mismatches are converted into force imbalances. A highly rigid system tends to translate small geometric deviations into larger force imbalances because little deformation is available to accommodate the mismatch. Conversely, a more compliant system can redistribute load through elastic deformation, reducing force concentration at the expense of increased displacement, tilting, or deflection. The numerical model therefore includes both gear-mesh stiffness and bearing stiffness. These stiffness contributions are combined into an effective branch stiffness that governs how geometric errors influence the final equilibrium solution.

Mesh Stiffness

Mesh stiffness represents the compliance of the tooth contacts between the sun-planet and ring-planet meshes. In the reduced-order model, the mesh stiffness is estimated from gear geometry rather than prescribed directly by the user. The sun-planet and ring-planet contacts are therefore represented as elastic elements with corresponding stiffnesses k_{SP} and k_{RP} .

For the sun-planet mesh, the transverse contact ratio is calculated as:

$$\epsilon_{\alpha,sp} = \frac{\sqrt{r_{a1}^2 - r_{b1}^2} - a_w \sin(\alpha_w) + \sqrt{r_{a2}^2 - r_{b2}^2}}{2\pi r_{b1}/z_1}$$

where ϵ_{α} represents the average number of tooth pairs simultaneously carrying load. A higher contact ratio generally increases the effective mesh stiffness because the transmitted load is shared across a larger contact region. The corresponding upper and lower stiffness bounds are then estimated as:

$$k_{sp,max} = \frac{\epsilon_{\alpha,sp} b}{\cos(\beta_b)} \cdot 1.3 \times 10^{10}, \quad k_{sp,min} = \frac{\epsilon_{\alpha,sp} b}{\cos(\beta_b)} \cdot 1.2 \times 10^{10}.$$

For the ring-planet mesh, the contact-ratio expression differs due to the internal-gear geometry of the ring engagement, although the stiffness formulation remains unchanged:

$$\epsilon_{\alpha,rp} = \frac{\sqrt{r_{a2}^2 - r_{b2}^2} + a_w \sin(\alpha_{w,krp}) - \sqrt{r_{a3}^2 - r_{b3}^2}}{2\pi r_{b2}/z_2}$$

$$k_{rp,\max} = \frac{\epsilon_{\alpha,rp} b}{\cos(\beta_b)} \cdot 1.3 \times 10^{10}, \quad k_{rp,\min} = \frac{\epsilon_{\alpha,rp} b}{\cos(\beta_b)} \cdot 1.2 \times 10^{10}$$

The face width b enters linearly into the stiffness formulation because a wider tooth provides a larger contact region over which the load can be distributed. The helix-angle correction $1/\cos(\beta_b)$ accounts for the inclined contact line present in helical gears and therefore modifies the effective stiffness along the line of action.

The upper and lower stiffness bounds are then used to construct the mean mesh stiffness and the amplitude of the stiffness variation:

$$k_{\text{mean}} = 0.5 (k_{\max} + k_{\min}), \quad k_{\text{amp}} = 0.5 (k_{\max} - k_{\min})$$

For the reference gear geometry used throughout the project, the contact ratio typically lies between approximately 1.4 and 1.7. This indicates that the mesh alternates between single-tooth and double-tooth contact during operation. When only one tooth pair is engaged, the mesh stiffness is lower; when two tooth pairs simultaneously share the load, the mesh stiffness increases. The quantities $k_{\max}$ and $k_{\min}$ therefore represent simplified bounds on the stiffness variation occurring throughout a mesh cycle.

Within the current version of the model, the mean stiffness k_{mean} is used to define the baseline branch compliance employed by the equilibrium solver. The amplitude k_{amp} quantifies the magnitude of the stiffness variation and forms the basis of the periodic mesh-stiffness representation introduced later in the phase-dependent analysis chapter. The empirical coefficients 1.2×10^{10} and 1.3×10^{10} are used to convert the geometric contact-ratio formulation into an absolute line-of-action stiffness expressed in N/m. These coefficients should be interpreted as reduced-order engineering approximations incorporating the combined influence of material elasticity and typical tooth-compliance effects. Their purpose is to provide realistic stiffness magnitudes without requiring the derivation of detailed tooth-level meshing relationships.

While mesh stiffness describes the compliance of the gear-tooth contacts, the flexibility of the planet support path also contributes to the overall branch compliance. This second contribution is represented in the model through the bearing stiffness formulation described in the following section.

Bearing Stiffness

Bearing stiffness k_B represents the compliance of the planet support path. In a planetary gear system, each planet is supported by a pin or shaft through a bearing, however this support is not perfectly rigid. Its flexibility influences how much a planet can move in response to geometric mismatch and therefore affects how strongly manufacturing errors are converted into load imbalance. A softer support path allows greater local displacement and can partially accommodate positional errors, whereas a stiffer support transfers these mismatches more directly into mesh forces.

The final tool uses five preset bearing-stiffness levels, summarized in Table 4.1.

Table 4.1: Bearing stiffness presets implemented in the numerical tool.

Bearing support condition	Preset Type	k_B [N/m]
Ball bearing with clearance	Highly compliant	1×10^6
Ball bearing without clearance	Compliant	1×10^7
Standard intermediate support	Baseline	5×10^7
Roller bearing with clearance	Stiff	1×10^8
Roller bearing without clearance	Highly stiff	1×10^9

These values should be interpreted as engineering presets rather than exact catalogue bearing stiffnesses. During the interim stage of the project, Palmgren-based bearing formulations were investigated to estimate bearing stiffness from bearing geometry and operating conditions. However, these relationships require detailed inputs such as rolling-element dimensions, contact geometry, preload conditions, and applied loads, which would have significantly increased the number of minimum user inputs. To preserve the simplicity of the tool while retaining realistic support-compliance effects, these representative order-of-magnitude stiffness levels were adopted instead.

The selected presets represent increasing support rigidity, from a highly compliant ball bearing with clearance to a relatively stiff roller bearing without clearance. Clearance refers to the free internal play that exists between the rolling elements and bearing races before load is transmitted. When clearance is present, the planet centre can move slightly before a significant restoring force develops, reducing the effective support stiffness. Removing clearance, for example through preload or tighter assembly, causes the bearing stiffness to engage immediately and therefore increases the rigidity of the support path.

The distinction between ball and roller bearings is represented through their different contact geometries. Ball bearings transmit load through point contacts, whereas cylindrical roller bearings transmit load through line contacts distributed over a larger area. Roller bearings are therefore generally stiffer and less compliant under load, which is reflected by the higher stiffness values assigned in the model.

Bearing stiffness is inherently difficult to define using a single universal value because it depends on many factors, including bearing size, internal geometry, preload, lubrication conditions, material properties, and operating load. The preset values are therefore intended for comparative and sensitivity studies rather than detailed bearing design. Their purpose is to allow the influence of support compliance on load-sharing behaviour to be investigated without requiring extensive bearing-specific data.

Effective Stiffness

The effective stiffness $k_{\text{eff},i}$ combines all compliance contributions associated with a single planet branch into one scalar value. This quantity is used directly within the equilibrium solver and represents the overall resistance of the branch to elastic deformation. For planet branch i , it is defined as:

$$k_{\text{eff},i} = \left[\frac{1}{k_{SP,i} + k_{RP,i}} + \frac{1}{k_{B,i}} \right]^{-1}$$

The sun-planet and ring-planet mesh stiffnesses, $k_{SP,i}$ and $k_{RP,i}$, are combined in parallel because both tooth contacts deform under the same branch displacement. The resulting mesh stiffness is then placed in series with the bearing stiffness $k_{B,i}$, since the bearing and mesh deflections accumulate along the load path between the planet gear and the carrier. This formulation allows each planet branch to be represented by a single stiffness value rather than tracking the individual mesh and support compliances separately. As a result, the equilibrium solver can evaluate the effect of geometric errors using one equivalent compliance parameter per planet branch while still retaining the influence of both tooth-contact and support flexibility.

A higher effective stiffness causes a given compatibility mismatch to generate larger force differences between planets, making the system more sensitive to manufacturing and assembly errors. Conversely, a lower effective stiffness allows greater elastic accommodation of these mismatches, promoting load redistribution and reducing force concentration between branches.

Stiffness Scaling and Optional Support

Mesh Scale Factor

The mesh scale factor is a user-defined multiplier applied to the calculated sun-planet and ring-planet mesh stiffnesses:

$$k_{SP} \leftarrow k_{SP} S_{\text{mesh}}, \quad k_{RP} \leftarrow k_{RP} S_{\text{mesh}}$$

The scaling is applied to both the mean mesh stiffness and the stiffness-variation amplitude. Consequently, it affects both constant-stiffness simulations and analyses that include periodic mesh-stiffness effects. The purpose of this parameter is not to replace the geometry-based stiffness calculation, but to account for the uncertainty associated with reduced-order mesh-stiffness models. Increasing the mesh scale factor produces a globally stiffer tooth-contact path, whereas decreasing it produces a more compliant mesh. This allows the influence of mesh-stiffness uncertainty on load-sharing behaviour to be investigated without modifying the underlying gear geometry.

Bearing Scale Factor

Similarly, the bearing scale factor is another user-defined multiplier, which gets applied to the selected bearing stiffness:

$$k_B \leftarrow k_B S_{\text{bearing}}$$

This parameter modifies the compliance of the planet support path while leaving the mesh stiffness unchanged. Since bearing stiffness depends on many factors, including bearing geometry, preload, clearance, lubrication conditions, and operating load, it is often difficult to define precisely without detailed bearing data. The bearing scale factor therefore provides a practical method for studying how uncertainty in support stiffness influences the effective stiffness of each planet branch. Increasing the factor produces a stiffer support path, while decreasing it increases support compliance.

This distinction is important because two systems may have the same effective stiffness but different physical reasons for that stiffness. A system may be limited by soft tooth contact, soft bearing support, or a combination of both. The GUI therefore separates mesh and bearing scale so that their individual effects on load sharing can be studied.

Optional Support Stiffness

The optional support stiffness parameters differ from the mesh and bearing scale factors because they do not modify the stiffness of individual planet branches. Instead, they introduce resistance to the motion of the floating sun gear itself. In the default configuration, the sun gear is treated as fully floating, allowing it to translate and tilt freely in response to the forces generated by the planet branches. Optional support stiffness can be introduced to represent a partially constrained support condition.

Two support terms are available: settlement support stiffness, k_w , which resists translational displacement of the floating sun; and tilt support stiffness, k_ϕ , which resists angular tilting of the floating sun. Increasing either support stiffness reduces the ability of the floating sun to self-centre in response to manufacturing and assembly errors. As a result, sun displacement and tilt are reduced, but the system also loses part of the load-sharing benefit associated with a freely floating configuration. These parameters are therefore intended for comparative studies in which the behaviour of a fully floating system is compared against partially constrained support conditions.

In practice, moderate support conditions are typically represented by values in the range $10^3 - 10^5$ N/m for the tilt support stiffness and $10^4 - 10^6$ N/m for the settlement support stiffness. Larger values progressively restrict the self-centering capability of the floating member, while values approaching $10^8 - 10^9$ N/m effectively behave as an artificial fixed support.

Thermal Stiffness Scaling

The second thermal effect included in the final model is a reduced-order correction to the mesh stiffness. The purpose of this correction is to represent the fact that temperature can change the effective stiffness of the gear mesh, without the complexity of incorporating a full thermo-elastic tooth-contact model.

The model defines the temperature difference as:

$$\Delta T = T - T_{\text{ref}}$$

where T is the operating temperature as input by the user and T_{ref} is the reference temperature. In the code, the default reference temperature is 20°C.

The stiffness correction is based on two simplified physical mechanisms. First, the elastic modulus of the gear material is assumed to decrease with increasing temperature, producing a gradual reduction in the effective mesh stiffness. Second, thermal expansion causes a small increase in the face width of the gear teeth, which slightly increases the effective contact region and therefore contributes positively to the mesh stiffness.

The combined thermal correction therefore reflects the competing influence of reduced material rigidity and increased contact geometry within a reduced-order framework.

The material stiffness reduction is represented using a first-order retained modulus ratio:

$$E_{\text{ratio}} = 1 - c_E \Delta T$$

where c_E is the temperature sensitivity coefficient representing the assumed rate at which retained mesh stiffness decreases with temperature:

$$c_E = 4 \times 10^{-4} \text{ } ^\circ\text{C}^{-1}$$

The second contribution is the thermal expansion of the face width. This is represented using the standard linear thermal expansion relation:

$$b_{\text{ratio}} = 1 + \alpha \Delta T$$

where α is the linear thermal expansion coefficient that had been previously introduced in Chapter 3. Since the reduced mesh stiffness model is assumed to scale approximately with both retained stiffness and face width, the final thermal stiffness scale is calculated as:

$$\text{scale} = E_{\text{ratio}} \cdot b_{\text{ratio}}$$

This scale is then applied to both the sun-planet and ring-planet mesh stiffnesses:

$$k_{SP}(T) = \text{scale} \cdot k_{SP}(T_{\text{ref}})$$

$$k_{RP}(T) = \text{scale} \cdot k_{RP}(T_{\text{ref}})$$

To prevent this simplified linear correction from producing unrealistic stiffness values, the final thermal stiffness scale is clamped between a minimum and a maximum value in the code. This means that the thermal correction is deliberately bounded and used only as a reduced-order modifier of mesh stiffness.

$$0.75 \leq \text{scale} \leq 1.05$$

Therefore, the thermal stiffness correction should be interpreted as an engineering approximation. It allows temperature to influence the mesh stiffness used in the load-sharing calculation, but it does not model detailed material behaviour, tooth contact deformation or thermal stress.

5. System Model and Assumptions

Model Overview

The numerical tool represents the planetary gearbox using a reduced-order equilibrium model developed specifically for load-sharing analysis. Rather than resolving full gear-tooth-contact mechanics or the structural behaviour of every component, the gearbox is represented through equivalent stiffnesses, compatibility errors, and the motion of a floating sun gear. This approach retains the dominant mechanisms governing load sharing while remaining computationally efficient. The assumptions used to construct the model are introduced in the following sections.

System Modelling Configurations

Floating Members in Planetary Systems

In planetary gear systems, a floating member is a component whose centre is not perfectly constrained and is therefore allowed to undergo small radial displacements under load. This additional freedom enables the system to partially self-correct manufacturing and assembly errors by redistributing the transmitted forces between planet branches. Although any of the principal members of a planetary gear set may theoretically be allowed to float, floating-sun configurations are among the most commonly studied arrangements because they provide effective self-centering behaviour and improved load sharing.

When the sun gear is allowed to move slightly from its nominal position, the forces transmitted through the planets generate an aligning behaviour that tends to reduce load imbalance and improve force distribution between branches. For this reason, the present model adopts a floating sun gear while treating the remaining members as geometrically constrained. The sun position is therefore not prescribed a priori but is obtained as part of the equilibrium solution, allowing the gear to settle at the position that balances the forces generated by the surrounding planet branches. From a modelling perspective, this configuration also provides a convenient way of coupling the planet branches through a small number of generalized coordinates while retaining the dominant load-sharing mechanisms.

Fixed Carrier Assumption

The planet carrier is treated as a rigid body, and carrier structural deformation is therefore neglected. This assumption is commonly adopted in reduced-order planetary-gear models

when the carrier stiffness is significantly larger than the stiffness associated with the gear meshes and support compliances.

Under this assumption, the carrier serves only as the geometric reference that defines the planet locations, while the load-sharing behaviour is governed primarily by the interaction between mesh stiffness, support compliance, geometric errors, and the motion of the floating sun gear. Neglecting carrier flexibility substantially reduces model complexity while retaining the mechanisms that have the greatest influence on load distribution.

Quasi-Static Analysis

The model is quasi-static, meaning that dynamic effects are neglected and the system is assumed to be in static equilibrium under the applied load at each evaluated configuration. Inertia, damping, vibration, resonance, and transient response are therefore not included. The response is governed only by compatibility conditions, stiffness relationships, and static equilibrium.

This does not mean that rotational effects are ignored entirely. The final model can evaluate the system over one full rotational cycle, from 0 to 2π , by introducing a phase variable γ . In this context, 'phase' represents the angular position of the system during the cycle, not physical time. The solver therefore depends on rotation angle γ , rather than on time t . At each phase step, the model updates the phase-dependent quantities, such as rotating eccentricity or periodic mesh stiffness, and then solves a new static equilibrium problem. The resulting forces, load-sharing factors, compatibility gaps, active contacts, stiffness values, and floating-sun displacement are recorded for that specific phase. The phase response is therefore a sequence of quasi-static equilibrium solutions, not a time-domain dynamic simulation.

This distinction is important because the system does not necessarily have one fixed load-sharing condition over a rotation. Rotating eccentricity and mesh-stiffness variation can cause the most-loaded planet and the severity of load imbalance to change with phase. However, because the model does not include inertia or damping, the phase sweep should be interpreted as a position-dependent quasi-static analysis rather than a prediction of vibration, dynamic amplification, or transient gearbox response.

Generalized Coordinates

Under the reduced-order assumptions of the model, the motion of the floating sun gear is represented using three generalized coordinates assembled into the vector

$$\mathbf{q} = \begin{bmatrix} w \\ \phi_x \\ \phi_y \end{bmatrix}$$

Rather than modelling the complete three-dimensional motion of every gearbox component, the solver represents the floating sun through a small set of rigid-body motions that capture the dominant mechanisms affecting load sharing. This reduction significantly decreases the computational complexity of the problem while retaining the ability to predict how manufacturing errors, assembly deviations and stiffness variations influence the distribution of load between planets.

The first coordinate, w , represents the settlement of the floating sun within the reduced-order equilibrium formulation. Physically, it corresponds to the overall approach of the floating member towards the planet branches under load. A positive settlement affects all planets equally and therefore represents the common displacement component experienced by the system.

The coordinates ϕ_x and ϕ_y represent small rigid-body tilts of the floating sun about two orthogonal axes. Unlike settlement, tilting does not affect all planets equally. Instead, it causes some planets to move closer to the sun while others move further away, depending on their angular position around the gearbox. These tilt degrees of freedom are therefore essential for describing how the floating member redistributes load in response to manufacturing errors, eccentricities and asymmetric stiffness conditions.

In addition, the model assumes that all displacements and rotations / tilts are small. This ensures linear kinematic relations and is consistent with the quasi-static, reduced-order framework. Geometric nonlinearities, such as large-deflection effects and change of contact angle with deformation, are therefore neglected.

Together, the settlement and tilt coordinates define the equilibrium position adopted by the floating sun during each quasi-static solution. Rather than remaining fixed at its nominal geometric centre, the sun is allowed to move to a new position that minimises the overall compatibility mismatch between the planet branches. This self-adjustment mechanism is one of the primary reasons why floating-member planetary gear systems exhibit improved load-sharing behaviour compared with fully constrained configurations.

Coupled Equilibrium Assumption

The final model treats all planet branches as mechanically coupled through the motion of the floating sun gear. As a result, the load carried by any individual planet cannot be determined independently from the remaining branches.

When a geometric error, stiffness variation, or positional deviation occurs on one planet branch, it modifies the equilibrium position of the floating sun. The resulting settlement and tilt of the sun then alter the compatibility conditions experienced by every other planet branch in the system. Consequently, a change affecting a single planet influences the force distribution throughout the entire gearbox rather than remaining a purely local effect.

Load sharing is therefore formulated as a global equilibrium problem in which all planet branches participate simultaneously in determining the final force distribution. The equilibrium position of the floating sun and the forces carried by the planets must be solved together as part of one coupled system. This assumption is fundamental to the present model because the self-adjusting motion of the floating member is the mechanism through which manufacturing errors, assembly deviations and stiffness variations are redistributed between planets.

Without this coupling effect, each planet could be analysed as an independent contact path and the load-sharing behaviour characteristic of floating-sun planetary gear systems would not be captured correctly.

Compression-Only Contact

Each planet branch is represented as a compression-only elastic contact. This means that a mesh contact may transmit compressive force, but cannot sustain tension. Physically, gear teeth can push against one another, but they cannot pull on each other once contact is lost. As a consequence, a planet branch only contributes to load transmission while its contact force remains positive. If the equilibrium solution predicts a negative contact force for a given branch, that planet is assumed to lose contact and no longer participates in carrying load. The corresponding contact force is therefore set to zero.

This assumption introduces the primary nonlinearity of the reduced-order model. The set of planets actively carrying load is not known in advance and depends on the equilibrium position adopted by the floating sun. For this reason, the active contact pattern must be determined as part of the solution process rather than being prescribed beforehand.

The compression-only assumption is essential for representing realistic load-sharing behaviour because any errors, component compliance differences and phase-dependent effects may cause some planet branches to engage more strongly than others. The numerical solver developed in the following chapter therefore determines both the equilibrium position of the floating sun and the set of active load-carrying planets simultaneously.

6. Numerical Solver

Solver Overview

The numerical solver that forms the mathematical basis behind the tool calculates the global displacement of the sun gear and the corresponding force carried by each planet branch. The formulation operates on the equations and assumptions that define the reduced-order planetary system model introduced in the previous chapters, consisting of N planet branches located at angular positions θ_i around a pitch radius r . Each branch is characterised by an equivalent compatibility error and an effective stiffness.

Starting from these quantities, the solver relates the motion of the floating sun to the local displacement experienced by each planet branch. The resulting branch deflections are converted into contact forces and assembled into a coupled equilibrium system. Because contact is assumed to be compression-only, the active set of load-carrying planets is not known in advance and must be determined as part of the solution process.

The following sections develop the mathematical formulation that transforms the assigned system errors and branch stiffnesses into the final equilibrium solution and load-sharing distribution.

Compatibility Formulation

Compatibility Gaps

Before the equilibrium problem can be assembled, the compatibility condition of each planet branch must be related to the motion of the floating sun gear. As discussed in Chapter 2, the combined geometric and manufacturing errors are first converted into compatibility gaps,

$$\delta_i = e_{\max} - e_i$$

where δ_i represents the relative gap that must be closed before planet branch i can fully engage in load transmission. These compatibility gaps act as the geometric inputs to the solver and are combined with the displacement of the floating sun to determine the elastic deflection and force carried by each planet branch. They therefore quantify the relative engagement difference between planet branches prior to elastic deformation.

Kinematic Mapping

The displacement of each planet contact is obtained from the generalized-coordinate representation of the floating sun motion in the vector form

$$\mathbf{q} = \begin{bmatrix} w \\ \phi_x \\ \phi_y \end{bmatrix}$$

For a planet located at angular position θ_i , the displacement produced by the floating sun at the corresponding mesh contact is obtained from rigid-body kinematics. The resulting contact displacement is

$$d_i = w + r\phi_x \sin \theta_i - r\phi_y \cos \theta_i$$

where r is the support radius used in the reduced-order model.

This expression shows that settlement contributes equally to all planet branches, while the tilt coordinates introduce position-dependent displacement variations around the circumference of the gearbox. As a result, the same floating-sun motion affects each planet differently depending on its angular location. For compact implementation, the displacement relation is written using the influence vector

$$\mathbf{v}_i = \begin{bmatrix} 1 \\ r \sin \theta_i \\ -r \cos \theta_i \end{bmatrix}$$

such that

$$d_i = \mathbf{v}_i^T \mathbf{q}$$

The influence vector defines how each generalized coordinate contributes to the displacement experienced by planet i . It is therefore the fundamental geometric quantity linking floating-sun motion to the individual planet branches.

Branch Deflection and Contact Force

Each planet branch is represented by a linear spring of effective stiffness k_i . Within the reduced-order formulation, this stiffness represents the combined compliance of the gear meshes and bearing support discussed in Chapter 4. The branch stiffness is treated as constant during each quasi-static equilibrium solve, although its value may vary between phase steps when phase-dependent stiffness effects are enabled.

Once the displacement generated by the sun gear is known, the elastic deflection of planet branch i is calculated as

$$\Delta_i = d_i - \delta_i = \mathbf{v}_i^T \mathbf{q} - \delta_i$$

The branch deflection consequently depends on both the compatibility gap associated with the planet and the displacement produced by the floating sun. A positive value of Δ_i indicates that the compatibility gap has been overcome and that the branch is capable of transmitting load. Assuming linear elastic behaviour, the corresponding contact force is obtained from Hooke's law:

$$F_i = k_i \Delta_i$$

Substituting the displacement relation from the previous section gives the final expression used in the solver for the force of each planet, F_i .

$$F_i = k_i (\mathbf{v}_i^T \mathbf{q} - \delta_i)$$

This expression combines the three key aspects of the reduced-order model: the geometric compatibility errors represented by δ_i , the floating-sun motion represented by $\mathbf{q}$, and the branch compliance represented by k_i . Together, these quantities determine the load carried by each planet branch.

At this stage, the force carried by an individual planet can be calculated for a given displacement state. However, the displacement state itself remains unknown because the floating-sun coordinates must satisfy equilibrium with all planet branches simultaneously. The determination of this equilibrium condition forms the basis of the global solver formulation.

Global Equilibrium Formulation

The planet branches cannot be solved independently. A local error or stiffness variation at one planet modifies the position of the sun gear, which in turn changes the elastic deflection of all other branches. The equilibrium problem must therefore be written as a coupled system in terms of the generalized displacement vector $\mathbf{q}$.

Energy Formulation

Under the quasi-static assumption, inertial effects are neglected and the equilibrium configuration is obtained from the stationary value of the total potential energy. Given the elastic deflection $\Delta_i = \mathbf{v}_i^T \mathbf{q} - \delta_i$ of an active planet branch i , the elastic strain energy stored in all active planet branches can then be expressed as:

$$U_b = \frac{1}{2} \sum_{i \in \mathcal{A}} k_i (\mathbf{v}_i^\top \mathbf{q} - \delta_i)^2$$

where $\mathcal{A}$ is the current set of active contacts and k_i is the effective stiffness of branch i . If an additional support stiffness is included, its contribution is written as

$$U_s = \frac{1}{2} \mathbf{q}^\top \mathbf{K}_s \mathbf{q}$$

From this, the total potential energy in the system can be given by:

$$\Pi = U_b + U_s - \mathbf{Q}_{\text{ext}}^\top \mathbf{q}$$

where $\mathbf{Q}_{\text{ext}}$ is the external generalized load vector. The elastic deflection of each planet branch is itself a function of the generalized displacement vector $\mathbf{q}$. Consequently, the strain energy stored in the branches – and therefore the total potential energy of the system – also depend on $\mathbf{q}$. Different displacement states correspond to different distributions of elastic deformation, giving rise to different values of total potential energy.

The equilibrium configuration of the system is obtained by finding the displacement state for which the total potential energy becomes stationary. Physically, this corresponds to the position adopted by the floating member when the elastic forces generated by the active planet branches exactly balance the applied external load. This condition is enforced by differentiating the total potential energy with respect to the generalized coordinates and requiring the resulting force residual to vanish:

$$\frac{\partial \Pi}{\partial \mathbf{q}} = \mathbf{0}$$

Which represents a state of static equilibrium. Subsequently, carrying out the differentiation transforms the energy formulation into an equivalent set of equilibrium equations. After collecting the resulting stiffness and forcing terms, the problem can be written in matrix form as:

$$\mathbf{K} \mathbf{q} = \mathbf{g}$$

In this expression, $\mathbf{K}$ represents the global stiffness of the active planet branches and optional support terms, while $\mathbf{g}$ combines the applied generalized load with the forcing effect created by the compatibility gaps. Solving this system gives the generalized displacement state of the floating sun.

Assembly of the Equilibrium System

The global stiffness matrix, $\mathbf{K}$, is assembled by summing the contribution of every active planet branch:

$$\mathbf{K} = \sum_{i \in \mathcal{A}} k_i \mathbf{v}_i \mathbf{v}_i^\top$$

remembering that k_i is the effective stiffness of any given branch i and $\mathbf{v}_i$ is the corresponding influence vector. Each branch therefore contributes a stiffness term that links the generalized displacement of the sun to the elastic response of the contact.

The generalized forcing vector, $\mathbf{g}$, is assembled as

$$\mathbf{g} = \mathbf{Q}_{\text{ext}} + \sum_{i \in \mathcal{A}} k_i \delta_i \mathbf{v}_i$$

where $\mathbf{Q}_{\text{ext}}$ represents the externally applied generalized load introduced earlier and the summation term accounts for the forcing effect introduced by the compatibility gaps. Each gap δ_i acts as a prescribed initial offset in planet branch i ; when multiplied by the branch stiffness k_i and projected through the influence vector $\mathbf{v}_i$, it produces an equivalent generalized forcing contribution. Manufacturing, assembly, thermal, and phase-dependent errors therefore influence the equilibrium solution through $\mathbf{g}$ rather than by directly modifying the generalized coordinates.

For the default floating-sun configuration, the equilibrium matrix is assembled solely from the active planet branches. However, the model also allows optional support stiffness terms to be introduced through

$$\mathbf{K}_{\text{support}} = \begin{bmatrix} k_w & 0 & 0 \\ 0 & k_\phi & 0 \\ 0 & 0 & k_\phi \end{bmatrix}$$

which modifies the global stiffness matrix to become

$$\mathbf{K} = \sum_{i \in \mathcal{A}} k_i \mathbf{v}_i \mathbf{v}_i^\top + \mathbf{K}_{\text{support}}$$

Here, k_w is the settlement support stiffness and k_ϕ is the tilt support stiffness. These additional terms do not alter the stiffness of the individual planet branches; instead, they introduce resistance to the global settlement and tilting motions of the floating sun. Physically, this can represent a partially constrained central member whose ability to self-centre is reduced: increasing these support stiffnesses reduces the freedom of the floating sun, which may reduce displacement but can also reduce the system's ability to compensate for manufacturing errors.

Once the stiffness matrix and forcing vector have been assembled, the generalized displacement state is obtained by solving the equilibrium system. The resulting coordinates are then used to recover the branch deflections and contact forces for each active planet.

Active Contact Resolution

The equilibrium system is first assembled using the current active contact set – assuming all planet branches remain in contact and contribute to the stiffness and forcing terms of the system – and solved for the generalized displacement vector. Once the equilibrium solution has been obtained, the resulting branch forces can be evaluated to verify whether this assumption remains physically admissible.

Since gear meshes can only transmit compressive force, any branch for which the calculated contact force becomes zero or negative must be considered inactive and removed from the active contact set. The compression-only contact condition is therefore

$$F_i \leq 0 \quad \implies \quad i \notin \mathcal{A}$$

where $\mathcal{A}$ denotes the active set of load-carrying planets.

When a planet leaves contact, its stiffness contribution is removed from the equilibrium system and the stiffness matrix and forcing vector are reassembled using only the remaining active branches. The equilibrium problem is then solved again to obtain an updated displacement state. This iterative procedure can be summarized as follows:

1. Initialise the active set with all planets engaged.
2. Assemble the equilibrium system using the current active set.
3. Solve for the generalized displacement vector $\mathbf{q}$.
4. Evaluate the branch forces F_i .
5. Remove any planets for which $F_i \leq 0$.
6. Repeat until all remaining planets satisfy $F_i > 0$.

This contact-update procedure introduces the only nonlinearity present in the numerical solver. For a fixed active set, the equilibrium problem remains linear. The nonlinearity arises solely from the possibility of planets entering or leaving contact as the equilibrium state changes. In most mechanical configurations all planets remain active throughout the solution. Contact loss generally occurs only for large compatibility gaps, severe manufacturing errors, or highly asymmetric system configurations.

Solver Sequence

The complete numerical solution procedure is summarized in the Figure 6.1 below.

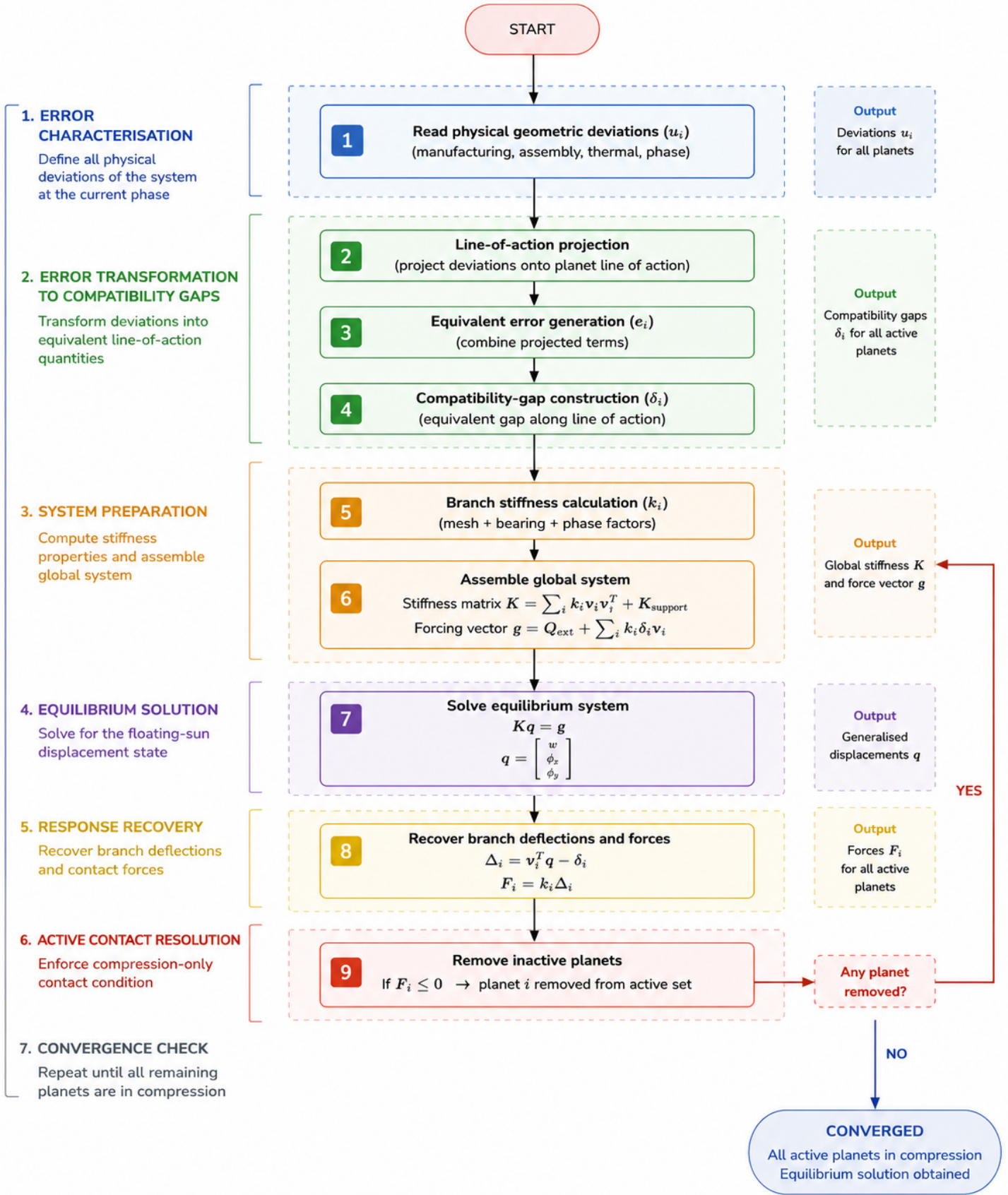


Figure 6.1: Execution workflow of the solver.

7. Phase-Dependent Load Redistribution

Why Phase-Dependent Analysis is Needed

The equilibrium formulation presented in the previous chapter determines the load-sharing state of the planetary system for a given set of stiffnesses and compatibility conditions. However, some physical mechanisms are inherently dependent on rotational position, and therefore cannot be represented by a single equilibrium solution. In particular, eccentricity and periodic mesh stiffness variation alter the effective geometry and compliance of the system as the gears rotate.

As a consequence, the load distribution between planets may change throughout a complete revolution even when the externally applied load remains constant. Evaluating the system at only one angular position may therefore overlook more severe loading conditions occurring elsewhere in the cycle. To capture these effects, the equilibrium problem must be solved repeatedly as the rotational position changes, allowing the variation in load sharing to be tracked over a complete revolution. This also then allows users of the numerical tool to be able to identify the most critical operating condition of the system, which may not coincide with the nominal or mean operating state solved initially by the equilibrium formulation.

Phase Sweep Concept

The phase-dependent behaviour of the system is evaluated by solving the equilibrium problem over a complete revolution of the phase angle,

$$0 \leq \gamma < 2\pi$$

At each phase step, the quantities affected by rotational position are updated according to the selected phase-dependent effects before the equilibrium system is assembled and solved. The resulting load-sharing metrics are then stored as functions of phase, allowing the response of the planetary system to be examined throughout the cycle.

It is important to note that the phase sweep does not constitute a dynamic simulation. The independent variable is the phase angle γ rather than physical time t , and each phase position is solved independently under the quasi-static assumption. No inertia, damping, vibration, resonance, or impact effects are included. Instead, the phase sweep

provides a computationally efficient method for evaluating how rotating errors and stiffness variations influence the quasi-static load distribution. The resulting phase histories can then be used to identify the most severe loading condition, referred to as the worst phase, as well as the variation of quantities such as planet force, load-sharing factor, and floating-member displacement throughout the cycle.

Periodic Eccentricity

Eccentricity represents a radial offset between the ideal centre of a rotating component and its actual axis of rotation. When the component rotates, this offset does not remain fixed in the global reference frame, but describes a circular path. As a result, the geometric mismatch seen by each planet branch varies with the phase angle γ .

In the model, the periodic eccentricity is represented by a rotating vector in the transverse plane:

$$\mathbf{e}_{\text{ecc}}(\gamma) = e_{\text{amp}} \begin{bmatrix} \cos(n_{\text{ecc}}\gamma + \gamma_0) \\ \sin(n_{\text{ecc}}\gamma + \gamma_0) \end{bmatrix}$$

where e_{amp} is the eccentricity amplitude, n_{ecc} is the eccentricity order, and γ_0 is the initial phase offset. If periodic eccentricity is disabled, this contribution is set to zero. For each planet branch, the rotating eccentricity is projected along the corresponding line-of-action direction. The scalar eccentricity contribution for planet i is therefore

$$e_{\text{ecc},i}(\gamma) = \mathbf{n}_{\text{LOA},i}^T \mathbf{e}_{\text{ecc}}(\gamma)$$

The projected eccentricity contribution is added to the equivalent-error framework before the compatibility gaps are updated. Thus, periodic eccentricity does not directly apply an additional force to the system. Instead, it modifies the compatibility gaps that determine which planet branches engage more strongly at each phase angle. As γ varies over the sweep, the most heavily loaded planet can therefore change even under constant applied torque.

Periodic Mesh Stiffness

In addition to geometric errors, the stiffness of a gear mesh is not strictly constant throughout a meshing cycle. As successive tooth pairs enter and leave contact, the number of teeth sharing the transmitted load changes, producing a periodic variation in mesh stiffness. This phenomenon is commonly referred to as time-varying mesh stiffness (TVMS).

A reduced-order representation of this behaviour was incorporated into the phase sweep model by allowing the sun-planet and ring-planet mesh stiffnesses to vary periodically with phase angle. The stiffness of each mesh is expressed as

$$k_{SP,i}(\gamma) = k_{SP,\text{mean}} + k_{SP,\text{amp}} \cos([n_{\text{TVMS}} \times \gamma] + \psi_i)$$

$$k_{RP,i}(\gamma) = k_{RP,\text{mean}} + k_{RP,\text{amp}} \cos([n_{\text{TVMS}} \times \gamma] + \psi_i)$$

where n_{TVMS} represents the selected harmonic order. In the present implementation, $n_{\text{TVMS}} = 1$, corresponding to one stiffness variation cycle over the phase sweep, however this variable can be changed by the user. The phase shift ψ_i introduces a staggered stiffness variation between planets, ensuring that each branch experiences the periodic stiffness cycle at a different angular position.

Unlike periodic eccentricity, which modifies the compatibility gaps through the equivalent error vector, periodic mesh stiffness affects the equilibrium solution by changing the stiffness contribution of each planet branch. Consequently, the load carried by each planet may vary throughout the phase sweep even when the compatibility conditions remain unchanged. It should be noted that the present implementation is intended to capture the first-order influence of mesh stiffness variation on load sharing. The model does not resolve individual tooth contacts or dynamic mesh excitation, but instead provides a computationally efficient representation suitable for quasi-static phase-dependent analysis.

Worst-Phase Identification

The objective of the phase sweep is not only to evaluate the variation of load sharing throughout a complete cycle, but also to identify the most severe operating condition experienced by the planetary system. At each phase step, the solver computes the system load-sharing factor,

$$K_\gamma = \frac{\max(F_i)}{\text{mean}(F_i)}$$

where F_i denotes the load carried by planet i . A value of $K_\gamma = 1$ corresponds to perfectly equal load sharing, while larger values indicate an increasingly uneven load distribution. The worst phase is defined as the phase angle for which the system load-sharing factor reaches its maximum value:

$$\gamma_{\text{worst}} = \underset{\gamma}{\operatorname{argmax}} (K_\gamma)$$

This phase represents the most critical loading condition encountered during the cycle, as it corresponds to the largest overload experienced by an individual planet relative to the mean transmitted load. For this reason, the worst phase is used throughout the tool as the reference condition for reporting final planet loads, load-sharing factors, and floating-sun displacements.

While the maximum value of K_γ identifies the most severe condition, the variation of load sharing throughout the cycle is also of interest. To quantify this sensitivity, the model computes the load-sharing span,

$$\Delta K_\gamma = \max(K_\gamma) - \min(K_\gamma)$$

which represents the difference between the most favourable and most severe load-sharing states observed during the phase sweep. A small value of ΔK_γ indicates that the load distribution remains relatively stable throughout the cycle, whereas a large value suggests a stronger dependence on phase angle and a greater influence of periodic effects such as eccentricity or mesh stiffness variation. The complete phase response is stored not only for K_γ , but also for individual planet LSF, force, and equivalent error values.

8. Output Metrics

The numerical solver provides a set of output metrics describing load distribution, floating-member motion, and contact state within the planetary system. To differentiate between these variables, their differences both mathematically and functionally are described in the following sections.

Planet Load Share

The individual planet load share represents the fraction of the total transmitted load carried by planet branch i :

$$\text{share}_i = \frac{F_i}{\sum_{j=1}^N F_j}$$

Under ideal symmetric conditions, each planet carries a nominal share of $1/N$. In the numerical tool, this value is reported for each planet as a decimal between 0 and 1 which then gets converted into a percentage, providing a direct measure of load distribution between planets. Inactive planets have a load share of zero and their load is redistributed among the remaining active branches. The total share remains conserved:

$$\sum_{i=1}^N \text{share}_i = 1.0 = 100\%$$

Load-Sharing Factor (LSF)

The load-sharing factor LSF_i compares the force carried by planet i with the average force that would be carried under ideal equal sharing:

$$\text{LSF}_i = \frac{F_i}{\frac{1}{N} \sum_{j=1}^N F_j}$$

Unlike the absolute force F_i , this metric is independent of the overall load level and therefore isolates the effect of geometry, stiffness, and contact conditions on load distribution. The interpretation is: if $\text{LSF}_i = 1.0$, the planet carries its ideal share of the total load. Having $\text{LSF}_i > 1.0$ means the planet is overloaded relative to the ideal average, and $\text{LSF}_i < 1.0$ denotes the planet is underloaded relative to the same average. The individual load-sharing factors also satisfy the system constraint:

$$\sum_{i=1}^N \text{LSF}_i = N$$

If a planet lifts off, its force and load-sharing factor are both set to zero. The remaining active planets must then carry the redistributed load, which appears as a corresponding rise in their LSF_i values. This makes the metric useful for identifying which branches become critical under unequal loading.

Global Load-Sharing Index

The global load-sharing index K_γ provides a system-level measure of the most severe instantaneous load imbalance. At each phase step, it is defined as the largest planet force normalized by the ideal average branch force:

$$K_\gamma = \frac{\max(F_i)}{\frac{1}{N} \sum_{j=1}^N F_j} = \max(\text{LSF}_i)$$

The maximum value of K_γ over the phase sweep represents the most severe load-sharing condition experienced by the system during the cycle. This is important because a gearbox may show acceptable average load sharing over a full rotation while still experiencing short intervals of severe overload. For physical gear systems, the peak value of K_γ identifies the planet branch and angular position most likely to govern fatigue, tooth root stress, and accelerated wear.

Floating-Sun Displacement

In addition to the load-sharing metrics, the tool reports the equilibrium position of the floating sun gear through the generalized displacement vector

$$\mathbf{q} = \begin{bmatrix} w \\ \phi_x \\ \phi_y \end{bmatrix}$$

where w is the settlement coordinate and ϕ_x and ϕ_y are the equilibrium tilt components. The settlement is reported in millimetres *mm*, while the tilt components are reported in radians *rad*. These quantities provide information about how the floating member adapts to compatibility variations within the system. Large values of settlement or tilt may indicate that the floating member is compensating for significant compatibility mismatches within the system.

Active Planet Status

Finally, the numerical tool also gives the contact status of each planet branch as an output. This quantity is represented as a binary indicator of a checked box, where the box being checked (✓) confirms active-load-carrying contact and unchecked symbolizing an inactive branch.

A planet is considered inactive when it does not participate in load transmission under the compression-only contact condition. Inactive planets therefore carry zero load and do not contribute to the equilibrium solution. This output provides a direct indication of contact loss within the planetary system and helps distinguish between ordinary unequal load sharing and more severe conditions involving planet lift-off.

9. Graphical User Interface

Main Screen



Figure 9.1: Main graphical user interface of the numerical tool.

Input Configuration

The input panel on the left tab allows the user to define the planetary gear geometry, operating conditions, bearing configuration, manufacturing errors, and optional advanced effects. Parameters are organised into logical groups to simplify model configuration and reduce the likelihood of inconsistent inputs.

Numerical Outputs

Following execution, the application reports the principal output metrics introduced in Chapter 8, including planet forces, load shares, load-sharing factors, floating-sun displacement, and active planet status. These can be found in the upper panel of the GUI screen. This section has two parts, one being a script giving a summary of all of the model variables and system parameters followed by the outputs, as well as a table beside it with information about how load has been distributed across the planets in more detail.

Graphical Visualization

Graphical outputs in the main section of the GUI provide a visual representation of the equilibrium solution, allowing rapid interpretation of load distribution, floating-member displacement, and phase-dependent behaviour. Particular emphasis was placed on presenting the numerical results in a form that can be interpreted rapidly without requiring direct inspection of the underlying data tables. Several visualizations were developed specifically for this purpose.

The first is the load-sharing factor (LSF) bar chart which can be seen in Figure 9.1, which displays the equilibrium load-sharing factor of each planet branch at the selected operating condition. The chart includes a reference line corresponding to the ideal value $LSF = 1$, allowing overloaded and underloaded planets to be identified immediately. This provides a direct visual indication of load imbalance within the system and highlights which planet branches are carrying a disproportionate share of the transmitted load.

The second is the error-component breakdown chart displayed below in Figure 9.2, which decomposes the equivalent compatibility error of each planet into its individual contributing sources. Manufacturing errors, profile modifications, runout, eccentricity, thermal effects, and other enabled contributions are displayed separately using a stacked representation, while the total equivalent error is shown simultaneously by the black line. This visualization allows the user to identify not only the magnitude of the final compatibility error but also the dominant physical mechanism responsible for it.

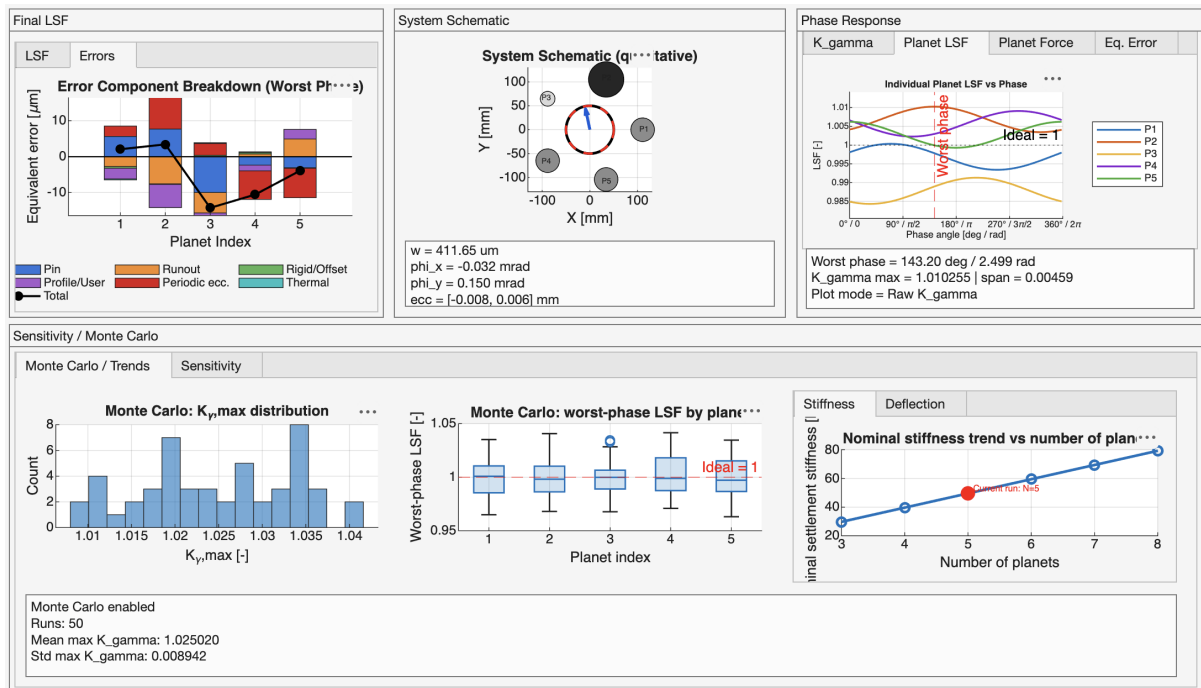


Figure 9.2: Visual / graphics section of the user interface

The phase-response panel was expanded to provide several complementary views of the system behaviour over one complete rotation. In addition to the global load-sharing index $K_\gamma(\gamma)$, the interface allows the user to inspect the individual planet load-sharing factors $\text{LSF}_i(\gamma)$ (visible in the Phase Response panel of Figure 9.2), contact forces $F_i(\gamma)$, and equivalent errors $e_i(\gamma)$ as functions of phase. This is useful because the worst global load-sharing condition does not necessarily reveal how each individual branch evolves throughout the cycle. By plotting the planet-level responses simultaneously, the user can identify which planets become critical, whether the most-loaded planet changes with phase, and how the equivalent error pattern drives the force redistribution.

Several display modes were also added for the global phase response. In raw mode, the plotted quantity is simply $K_\gamma(\gamma)$. In normalized mode, the response is scaled by its mean value over the cycle:

$$K_{\gamma,\text{norm}}(\gamma) = \frac{K_\gamma(\gamma)}{\overline{K_\gamma}}$$

where $\overline{K_\gamma}$ is the mean value of K_γ over the phase sweep, given by the equation

$$\overline{K_\gamma} = \frac{1}{N_\gamma} \sum_{j=1}^{N_\gamma} K_\gamma(\gamma_j)$$

The deviation-from-mean mode plots

$$\Delta K_\gamma(\gamma) = K_\gamma(\gamma) - \overline{K_\gamma}$$

while the percentage-variation mode plots

$$K_{\gamma,\%}(\gamma) = 100 \times \frac{K_\gamma(\gamma) - \overline{K_\gamma}}{\overline{K_\gamma}}$$

These alternative views do not modify the numerical solution; they only change how the same phase-response data are displayed. They were included to make the phase sensitivity of the system easier to interpret, especially when the absolute variation in K_γ is small but still relevant for comparing different configurations. They are available for selection in the *Advanced Effects* section in the Phase Response dropdown.

Advanced Studies

Additional plots are provided for uncertainty quantification and parameter studies. Monte Carlo analysis evaluates the statistical sensitivity of the system to manufacturing variations, while the sensitivity module investigates the influence of selected parameters on the worst-case load-sharing condition. Lastly, nominal stiffness and deflection per number of planets are also included for analysis of how the system would behave in respect to its assigned parameters for a higher or lower number of planets. These plots are located at the bottom panel between the two tabs.

10. Verification and Validation Strategy

Purpose

Verification was performed to confirm that the numerical tool reproduces known physical behaviours and published trends before being used for exploratory studies. Two verification cases were selected. The first checks whether the solver reproduces the expected perfect load-sharing condition of a symmetric floating planetary system, otherwise referred to as the Ideal Configuration. The second compares the predicted response to the experimental observations reported by Ligata et al. [1] for a planetary gear set containing a controlled manufacturing error.

Symmetric Zero-Error Case

Before applying the numerical tool to configurations containing manufacturing errors or other sources of asymmetry, a verification case was performed using an ideal planetary system. The purpose of this test was to confirm that the numerical implementation itself does not introduce artificial load imbalance when the physical system is perfectly symmetric.

The verification case consists of a three-planet configuration with all manufacturing and assembly errors set to zero, also referred to as the Ideal Configuration relating to the Translational Analogy framework [2]. Periodic eccentricity, thermal mismatch, and other optional perturbations were also disabled. Under these conditions, every planet branch is geometrically identical, possesses the same stiffness, and occupies an equivalent angular position within the planetary system.

From a mechanical perspective, perfect load sharing is expected. Since no planet possesses any geometric or stiffness advantage over the others, there is no mechanism capable of causing unequal load distribution. The equilibrium solution must therefore remain fully symmetric, resulting in identical force transmission through all planet branches. Consequently, the expected outputs are

$$F_1 = F_2 = F_3$$

$$\text{share}_1 = \text{share}_2 = \text{share}_3 = \frac{1}{3}$$

$$\text{LSF}_1 = \text{LSF}_2 = \text{LSF}_3 = 1$$

$$K_\gamma = 1$$

In addition, the floating sun remains in its nominal equilibrium position, with zero tilt and displacement due to the absence of any compatibility mismatch requiring compensation.

The numerical results shown in Figure 10.1 reproduce this behaviour exactly. All three planets carry identical loads, the load-sharing factors remain equal to unity, and the global load-sharing index satisfies $K_\gamma = 1$. No artificial asymmetry is introduced by the solver, demonstrating that the equilibrium formulation, force recovery procedure, and load-sharing calculations are internally consistent. This verification provides confidence that any load imbalance observed in subsequent analyses originates from the physical effects introduced into the model, such as manufacturing errors, stiffness variations, thermal effects, or periodic eccentricity, rather than from numerical errors within the implementation itself.

Planet	Angle_deg	Force_N	Share_pct	LSF	Active	EqError_um
1	0	491.4246	33.3333	1.0000	☒	0
2	120.0000	491.4246	33.3333	1.0000	☒	0
3	240.0000	491.4246	33.3333	1.0000	☒	0

Figure 10.1: Output table of the numerical tool for the zero-error ideal symmetry verification case

An additional visual verification is provided by the planetary-system schematic displayed in Figure 10.2 within the graphical user interface. In this representation, the colour intensity of each planet is linked directly to the calculated load carried by that branch, with darker tones indicating higher loads and lighter tones indicating lower loads. Since the ideal verification case produces identical forces on all three planets, the corresponding schematic displays all planets with the same shade of grey. This provides a qualitative confirmation that the load distribution is perfectly balanced and that no planet is carrying more load than another.

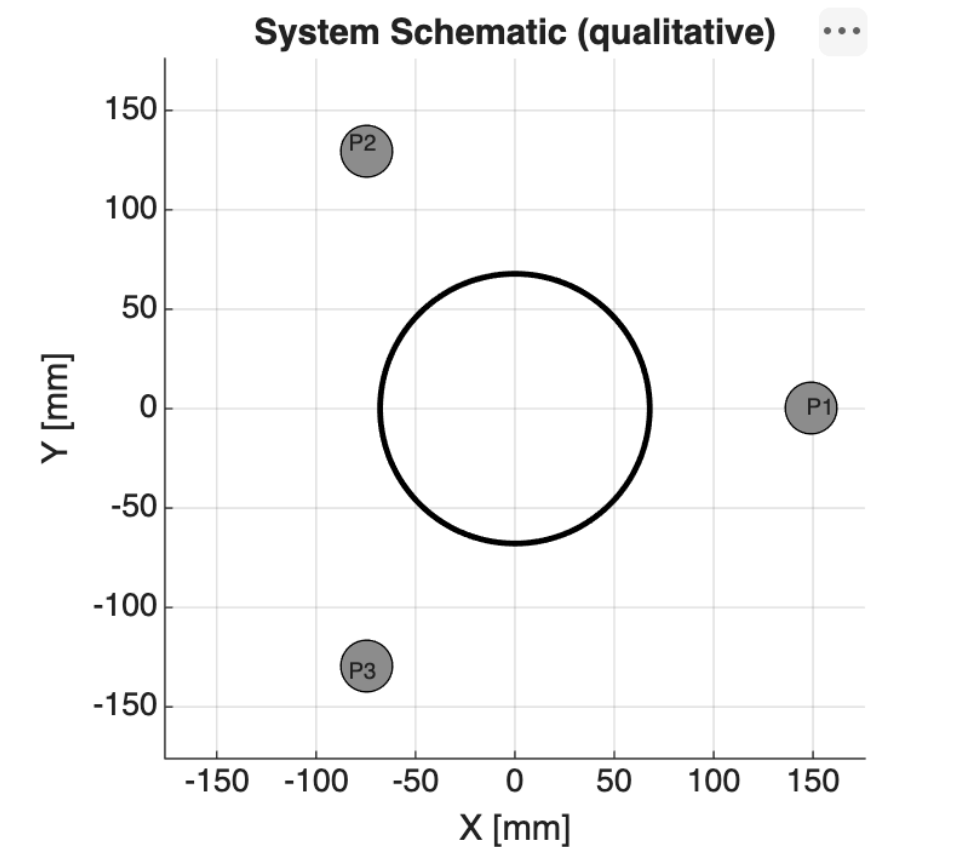


Figure 10.2: Qualitative schematic of the 3-planet system

Literature-Based Comparison

Reference Parameters

The parameter set reported by Ligata et al. [1] was adopted as the reference configuration used throughout the development and verification of the numerical tool. In addition to serving as a literature benchmark, these values were implemented as the default geometric parameters within the graphical user interface whenever user-specific gearbox information is unavailable.

This decision was made to reduce the number of mandatory inputs required from the user while still providing a physically representative planetary gear system on which the analysis can be performed immediately. As a result, the more detailed geometric quantities are treated as advanced settings that can be modified when a specific gearbox configuration is known, while the default values provide a practical baseline for exploratory studies, sensitivity analyses, and educational use of the tool. The original reference parameter set reported by Ligata et al. is reproduced in Figure 10.3 for completeness and to provide the basis of the default configuration implemented within the numerical tool.

Parameter	Sun	Planet	Ring
Number of teeth	73	26	125
Normal module	1.81	1.81	1.81
Pressure angle, deg	23.04	23.04	
Helix angle, deg	13.12	13.12	
Center distance	92.12		
Active face width	25	25	

Figure 10.3: Original gear geometry of the system used in the paper *An Experimental Study of the Influence of Manufacturing Errors on the Planetary Gear Stresses and Planet Load Sharing*, H. Ligata, 2008

Results Summary

The literature-based verification case was based on the experimental study that had been conducted in this same paper, in which four-, five-, and six-planet carrier configurations were tested with a controlled tangential pinhole position error applied to Planet 1. The reference study considered error magnitudes of $0\ \mu\text{m}$, $35\ \mu\text{m}$, and $70\ \mu\text{m}$, allowing the influence of a known manufacturing deviation to be isolated from the otherwise symmetric gear configuration.

In the present verification, the numerical tool was configured using the same reference gear parameters and a single-planet error case of $-70\ \mu\text{m}$ on Planet 1 for torques of 600 Nm, 800 Nm, and 1000 Nm. The objective was not to reproduce every experimental strain measurement exactly, but rather to verify that the solver reproduces the same load-sharing behaviour as had been reported for the system in the reference study.

For the four-planet system, the model predicts the characteristic paired behaviour of diametrically opposed planets. For the five-planet system, the imposed error causes one planet to become preferentially loaded while the adjacent planets provide the primary compensating response, leaving the first planet P_1 with the smallest load share. The six-planet configuration exhibits the greatest sensitivity to the same manufacturing error, producing the largest deviations from ideal load sharing ($1/6$ of the load share would give a value of 0.167 in ideal configurations). The tables below compare the planet load-share values reported by the paper with those obtained from the numerical tool.

Table 10.1: Comparison of planet load-share values for the four-planet configuration with a $-70 \mu m$ Planet 1 error.

Planet	600 Nm		800 Nm		1000 Nm	
	Paper	Tool	Paper	Tool	Paper	Tool
P1	0.14	0.16	0.18	0.19	0.18	0.19
P2	0.33	0.34	0.32	0.31	0.31	0.31
P3	0.12	0.15	0.12	0.18	0.16	0.19
P4	0.41	0.34	0.38	0.33	0.35	0.31

Table 10.2: Comparison of planet load-share values for the five-planet configuration with a $-70 \mu m$ Planet 1 error.

Planet	600 Nm		800 Nm		1000 Nm	
	Paper	Tool	Paper	Tool	Paper	Tool
P1	0.02	0.05	0.07	0.09	0.07	0.11
P2	0.34	0.32	0.29	0.29	0.29	0.27
P3	0.16	0.16	0.16	0.16	0.17	0.17
P4	0.13	0.15	0.16	0.17	0.16	0.17
P5	0.35	0.32	0.32	0.29	0.31	0.28

Table 10.3: Comparison of planet load-share values for the six-planet configuration with a $-70 \mu m$ Planet 1 error.

Planet	600 Nm		800 Nm		1000 Nm	
	Paper	Tool	Paper	Tool	Paper	Tool
P1	0.02	0.00	0.01	0.02	0.02	0.05
P2	0.25	0.28	0.26	0.26	0.26	0.24
P3	0.14	0.17	0.14	0.17	0.14	0.17
P4	0.13	0.11	0.12	0.12	0.12	0.13
P5	0.20	0.17	0.20	0.17	0.20	0.17
P6	0.26	0.27	0.27	0.26	0.26	0.24

For each comparison point, the absolute deviation between the reference value and the numerical tool output was calculated as

$$e_{i,T} = \left| L_{i,T}^{\text{paper}} - L_{i,T}^{\text{tool}} \right|$$

where $L_{i,T}^{\text{paper}}$ is the planet load-share value reported in the reference paper and $L_{i,T}^{\text{tool}}$ is the corresponding value obtained from the numerical tool for planet i at torque level T . The mean absolute deviation over all comparison points was then calculated as

$$\text{MAE} = \frac{1}{N_{\text{tot}}} \sum_T \sum_i e_{i,T}$$

with $N_{\text{tot}} = 45$ total load-share values.

Across all 45 load-share values extracted from the reference study, the numerical tool produced a mean absolute deviation of approximately 0.013 load-share units, corresponding to an average difference of roughly 1.3 percentage points per planet. The maximum observed deviation was approximately 0.07. Given the simplified quasi-static formulation of the model and the experimental nature of the reference data, this level of agreement can be considered reasonably competent. More importantly, it provides numerical evidence that the numerical tool can reproduce the same load-sharing trends and sensitivity to manufacturing errors as those that were recorded in the literature, confirming that it captures the dominant mechanisms governing manufacturing-error-induced load sharing behavior in planetary gear systems.

11. Advanced Analyses

Sensitivity Analyses

Purpose

A single simulation provides information about the behaviour of a planetary system for one particular set of parameters. However, it does not indicate which modelling assumptions or design variables have the greatest influence on load sharing. Sensitivity analysis was therefore incorporated as a feature of the numerical tool to quantify how variations in selected parameters affect the predicted system response. By varying one parameter at a time while keeping all others fixed, the influence of individual modelling assumptions can be isolated and compared. Throughout this study, the maximum load-sharing index $K_{\gamma, \max}$ is used as the primary response metric, allowing the relative importance of different parameters to be assessed consistently.

The sensitivity framework focuses on three parameters that are expected to have a significant influence on load sharing: periodic eccentricity amplitude, mesh stiffness, and bearing stiffness. These quantities were selected because they represent three distinct mechanisms affecting the equilibrium solution. Eccentricity modifies the compatibility conditions between planets, mesh stiffness controls the relationship between compatibility errors and transmitted force, and bearing stiffness influences the ability of the floating member to redistribute load through self-centering.

By varying each parameter independently, the analysis can identify which aspects of the system have the greatest influence on the predicted load-sharing behaviour. The objective is not to determine whether a parameter is beneficial or detrimental, but rather to quantify how strongly variations in that parameter influence the predicted load-sharing response.

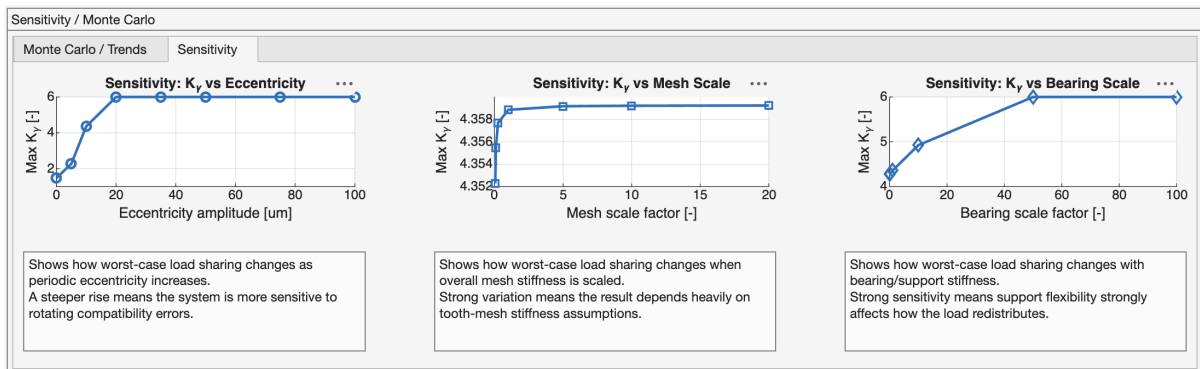


Figure 11.1: Sensitivity Studies output panel in the numerical tool GUI screen

Eccentricity Sensitivity

Periodic eccentricity introduces a phase-dependent compatibility error that varies continuously throughout one rotation of the planetary system. Although the eccentricity formulation itself has already been described in the phase-sweep analysis, its amplitude can have a significant influence on the severity of the resulting load imbalance. To quantify this effect, a sensitivity study was added in which the eccentricity amplitude was varied while all other model parameters remained unchanged. For each eccentricity level, the complete phase-sweep procedure is executed and the corresponding maximum load-sharing index $K_{\gamma,\max}$ extracted.

Increasing eccentricity amplitude increases the magnitude of the periodic compatibility mismatch experienced by the planet branches. As a result, the floating member must undergo larger positional adjustments in order to maintain equilibrium, potentially producing greater variation in the load-sharing response throughout the cycle. This effect may appear as an increase in $K_{\gamma,\max}$, a shift in the location of the worst phase, or, in sufficiently severe cases, the loss of contact of one or more planets.

The purpose of this study is therefore to evaluate the sensitivity of the planetary system to rotating geometric deviations and to determine whether eccentricity acts as a dominant driver of load-sharing variation. A steep increase in $K_{\gamma,\max}$ with eccentricity amplitude would indicate that the system is particularly sensitive to rotating compatibility errors, whereas a weak response would suggest that the floating mechanism is capable of effectively compensating for such disturbances. In the present implementation, the eccentricity amplitude is set to vary between $5\ \mu m$, $10\ \mu m$, and $20\ \mu m$, corresponding to low, medium, and high excitation levels respectively.

These values were selected to provide a simple comparison between mild, moderate, and severe periodic compatibility disturbances while remaining representative of the order of magnitude typically encountered in practical manufacturing and assembly deviations.

Mesh Stiffness Study

Mesh stiffness governs the relationship between compatibility mismatch and transmitted force within each planet branch. For a given geometric error pattern, a stiffer mesh produces larger force variations for the same relative displacement, whereas a more compliant mesh allows a greater degree of deformation before significant load concentration occurs. To evaluate this effect, a sensitivity study can be performed in which the mesh stiffness varies while all other model parameters are held constant. For each stiffness level, the corresponding $K_{\gamma,\max}$ response is evaluated.

From a physical perspective, increasing mesh stiffness does not necessarily improve load sharing. A stiffer system may reduce overall displacement while simultaneously

becoming more sensitive to compatibility errors, resulting in larger force concentrations on individual planets. Conversely, a more compliant system may tolerate geometric mismatch more effectively through elastic redistribution, although this typically occurs at the expense of increased floating-member displacement. The objective of this study is therefore to evaluate the trade-off between compliance and force concentration and to determine the influence of mesh stiffness on the overall load-sharing behaviour of the planetary system.

Bearing Stiffness Study

Bearing stiffness influences the compliance of the support path connecting each planet branch to the carrier structure. As bearing stiffness changes, the ability of the planetary system to accommodate geometric incompatibilities through elastic deformation also changes. To assess this effect, bearing stiffness is varied while all other parameters remain unchanged. The resulting equilibrium solutions can then be compared using the maximum load-sharing index $K_{\gamma,\max}$.

In general, softer bearing supports provide additional compliance within the load path and may assist the self-centering behaviour of the floating member. Conversely, stiffer supports reduce elastic accommodation and may increase the sensitivity of the system to manufacturing and assembly errors. As with mesh stiffness, the relationship is therefore not strictly beneficial or detrimental; instead, it reflects a trade-off between displacement and load concentration.

Within the numerical tool, both mesh stiffness and bearing stiffness sensitivity studies are implemented through dimensionless scaling factors applied to the baseline model. The mesh and bearing scales are varied according to

$$S_{\text{mesh}}, S_{\text{bearing}} = [0.70, 0.85, 1.00, 1.15, 1.30]$$

representing systems that are globally softer or stiffer than the reference configuration. The baseline case corresponds to a scale factor of 1.00, for which the calculated mesh stiffnesses and selected bearing stiffness presets are used without modification.

Monte Carlo Analysis

Purpose

Manufacturing and assembly errors are inherently uncertain and cannot be represented adequately by a single deterministic configuration. Consequently, a Monte Carlo framework was implemented to evaluate the effect of manufacturing variability on load-sharing

performance. Rather than analysing one specific error pattern, the solver generates multiple random realizations of the selected error model. Each realization represents a possible manufactured gearbox and is analysed independently using the complete phase-sweep procedure. The resulting collection of solutions provides a statistical description of the expected system behaviour and allows the robustness of the design to be assessed.

Randomized Error Generation

For each Monte Carlo realization, a new set of manufacturing errors is generated according to the selected tolerance assumptions. To ensure that each realization produces a different error pattern while remaining reproducible, the random-number generator is seeded incrementally according to

$$\text{seed}_{\text{run}} = \text{seed}_{\text{base}} + r - 1$$

where r denotes the realization number. This approach allows the entire study to be repeated consistently while ensuring that each run represents a distinct manufacturing configuration. All geometric parameters, stiffness parameters, and solver settings remain fixed throughout the Monte Carlo study. Only the manufacturing-error realization changes between runs. Consequently, the resulting variation in load-sharing behaviour can be attributed directly to manufacturing uncertainty rather than to changes in the underlying model configuration. Each realization performs the complete phase-sweep procedure and extracts the corresponding worst-phase response. The resulting collection of worst-case solutions forms the basis of the subsequent statistical analysis.

Statistical Outputs

The Monte Carlo framework produces a statistical description of the system response rather than a single deterministic result. For each realization, the solver records the principal output metrics, including the maximum load-sharing index $K_{\gamma, \text{max}}$, planet load-sharing factors, floating-sun displacement, and worst-phase location. From these data, statistical quantities such as the mean value, standard deviation, and distribution of $K_{\gamma, \text{max}}$ can be evaluated. These metrics provide information not only about the expected behaviour of the system, but also about its sensitivity to manufacturing variability.

A narrow distribution of $K_{\gamma, \text{max}}$ indicates that the planetary system responds consistently to the assumed tolerance range, suggesting a robust design that is relatively insensitive to small manufacturing variations. Conversely, a wide distribution implies that small changes in the error realization can significantly alter the load-sharing behaviour, indicating a greater sensitivity to manufacturing uncertainty.

Particular attention should be paid to the upper tail of the distribution. Although such cases may occur infrequently, they correspond to the more severe combinations of manufacturing errors and therefore represent the loading conditions most likely to govern design safety. For this reason, the Monte Carlo analysis complements the deterministic studies by providing information not only about nominal performance, but also about the variability and reliability of the predicted response.

12. Limitations and Future Improvements

The numerical tool developed in this work intentionally adopts a reduced-order quasi-static formulation in order to provide rapid load-sharing predictions with modest input requirements. While this approach offers significant computational advantages, it also introduces a number of limitations that define the range of situations for which the model can be expected to provide reliable results. The principal limitations of the current implementation, together with potential future extensions, are discussed in the following sections.

Experimental Validation

The numerical tool was developed and verified using theoretical test cases and published experimental data. Although these studies demonstrate that the implementation reproduces expected load-sharing behaviour, the model has not been validated against a dedicated experimental gearbox for which the manufacturing errors, operating conditions, and planet loads are known precisely.

Due to this, the present work should be interpreted as a physically motivated predictive tool rather than a fully validated representation of a specific gearbox. The predicted load-sharing trends, sensitivity studies, and phase-dependent behaviours are expected to be representative of real systems, but their quantitative accuracy cannot be assessed without experimental measurements. A natural extension of this work would be the development of a dedicated test rig or the acquisition of industrial gearbox data, as well as finite element analysis to compare the real system behaviour in parallel with a predictive model of a higher degree of mechanical simulation accuracy. Direct measurements of planet loads, carrier displacement, or strain responses would also provide a basis for evaluating the accuracy of the numerical tool and refining its assumptions where necessary.

Dynamic Effects

The most significant limitation of the present model, however, is arguably the quasi-static assumption adopted throughout the formulation. The equilibrium solution is obtained by balancing elastic forces at each phase position independently, meaning that inertial and damping effects are neglected. Consequently, the model cannot capture dynamic phenomena such as resonance, vibration amplification, transient responses, or dynamic load factors, which play a significant role in the behaviour of real epicyclic gear systems.

Although the phase-sweep procedure introduces periodic variations in compatibility conditions and mesh stiffness, it cannot be interpreted as a true time-domain dynamic simulation. The phase angle acts only as a parametric variable used to evaluate a sequence of equilibrium states. Therefore, the model predicts how load sharing varies throughout a cycle, but not how the system dynamically responds while rotating.

For many engineering applications, this simplification is acceptable because the dominant drivers of load imbalance are geometric errors, stiffness asymmetries, and compatibility mismatches, all of which are captured by the present formulation. However, as rotational speed increases, dynamic effects become increasingly important and may produce load concentrations that exceed the quasi-static predictions. A natural extension of the current framework would be the reintroduction of kinetic energy into the Lagrangian formulation, resulting in a system of second-order differential equations of the form

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}(t)\mathbf{q} = \mathbf{f}(t)$$

where $\mathbf{M}$ is the inertia matrix, $\mathbf{C}$ is a damping matrix, and $\mathbf{K}(t)$ contains the time-varying stiffness terms. Such an extension would allow the prediction of resonance behaviour, dynamic load amplification, and speed-dependent load-sharing effects while preserving the reduced-order philosophy of the current model.

Carrier Flexibility

The present formulation assumes that the carrier behaves as a rigid body. This simplification allows the equilibrium problem to be represented using only the three generalized coordinates associated with the floating sun, significantly reducing the complexity of the model. In practice, however, real carriers possess a finite structural stiffness and may undergo local deformation under load. Carrier-arm bending, web flexibility, and support compliance can all influence the distribution of load between planets, particularly in lightweight or large-diameter planetary gearboxes.

A possible improvement of the current framework would be the incorporation of carrier flexibility through compliance superposition. In this approach, a carrier compliance matrix could be obtained from an offline finite element model and incorporated into the reduced-order formulation without fundamentally altering the structure of the solver. This would preserve the computational efficiency of the numerical tool while allowing the influence of carrier deformation on load-sharing behaviour to be represented more realistically.

Helical and Multi-Stage Gearboxes

The present numerical tool was developed primarily for a single planetary stage and focuses on the axial compatibility relationships governing load sharing between planets. While this is sufficient for the spur-gear systems considered throughout this work, additional complexities arise in helical and multi-stage gearboxes.

Helical gears introduce coupled force components and additional sources of misalignment, while multi-stage transmissions require the interaction between multiple planetary sets to be represented simultaneously. Extending the current model to these configurations would require modified influence vectors, additional coupling terms, and potentially an expanded set of generalized coordinates involving more degrees of freedom. Nevertheless, the reduced-order framework developed in this project provides a suitable foundation upon which such advanced concepts could be incorporated in future versions of the numerical tool.

Advanced Thermal Effects

Thermal effects are included in the present model through simplified thermal mismatch and stiffness-scaling formulations. These additions allow the numerical tool to investigate how changes in operating temperature may influence compatibility conditions and load-sharing behaviour. As a result, the model is capable of capturing first-order thermal trends and comparing the relative influence of different thermal scenarios on the equilibrium solution.

However, the current implementation does not constitute a full thermo-mechanical analysis. Temperature is treated as a prescribed input rather than a quantity that evolves as a consequence of operating conditions. The model does not resolve temperature distributions within the gears, bearings, or housing, nor does it account for transient heating, heat transfer mechanisms, or thermo-elastic contact effects at the tooth level. Consequently, the thermal outputs should be interpreted as comparative indicators of system behaviour rather than precise predictions of thermal stresses or temperatures.

A possible future improvement would be the coupling of the load-sharing solver with a dedicated thermal model capable of predicting temperature fields throughout the gearbox. Inputs of specific gear material could also serve as a valuable input to include a broader range of relevant metal heat coefficients. Such an approach would allow thermal expansion, stiffness variation, and load sharing to be solved in a more physically consistent manner while retaining the reduced-order philosophy of the present framework. This would be particularly valuable for applications operating under high loads, elevated temperatures, or significant thermal gradients.

13. Conclusion

The objective of this project was to develop a numerical tool capable of studying load-sharing behaviour in epicyclic gear systems while maintaining a balance between physical realism, computational efficiency, and ease of use. Particular emphasis was placed on understanding how manufacturing errors, stiffness variations, and floating-member motion influence the distribution of load between planet gears.

To achieve this objective, a quasi-static reduced-order model was developed in which a given planetary gear system is represented through equivalent stiffnesses, compatibility gaps, and a floating sun gear described by a small set of generalized coordinates. The final formulation solves the coupled equilibrium of all planet branches simultaneously, allowing local geometric or stiffness variations to influence the global force distribution throughout the system. Additional capabilities were incorporated as the scope of the numerical tool progressed, including thermal modelling, phase-dependent effects, active contact resolution, sensitivity analyses, randomization studies, and a graphical user interface for visualization and interpretation of results.

The resulting final version of the tool is able to predict planet-by-planet forces, load-sharing factors, floating-sun displacement, active contact status, and worst-phase loading conditions for a wide range of planetary gear configurations. Verification cases demonstrated that the model reproduces the expected behaviour for both ideal symmetric systems and real test cases that had been studied in literature, providing a measurable scope of precision in the simulation capabilities of the model and solver.

Although the model does not attempt to replace high-fidelity finite element or dynamic simulations, it successfully captures the dominant mechanisms governing load sharing while retaining low computational cost. This makes it particularly well-suited to preliminary design studies, parameter investigations, sensitivity analyses, and educational applications where rapid evaluation of epicyclic gear system behaviour is required.

Future developments could further improve the accuracy of the model through the inclusion of dynamic effects, carrier flexibility, more advanced thermal formulations, and additional validation against experimental measurements or finite element simulations. Nevertheless, the present work demonstrates that a reduced-order equilibrium approach can provide a practical and physically meaningful framework for studying load-sharing behaviour in planetary gear systems while remaining accessible to engineering users.

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